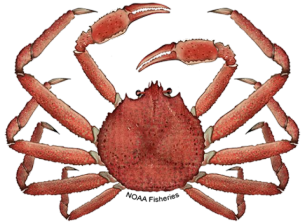
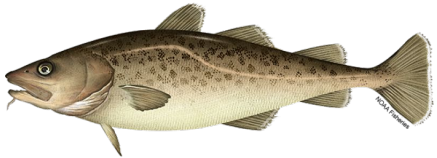
Crab and cod ‘Omics projects update for Auke Bay Genetics group

Laura Spencer, NOAA AFSC Affiliate at Univ. of Washington





AFSC Kodiak
Chris Long
Jennifer Gardner

AFSC Newport
Ben Laurel
Mary Beth Rew-Hicks
Emily Schlesinger
Tom Hurst

AFSC Seattle
Ingrid Spies

UW Seattle
Steven Roberts

June 15th, 2026

Collaborations with NOAA experimental laboratories









NOAA Newport Lab,
Behavioral Ecology Program





NOAA Kodiak Lab,
Shellfish Assessment Program

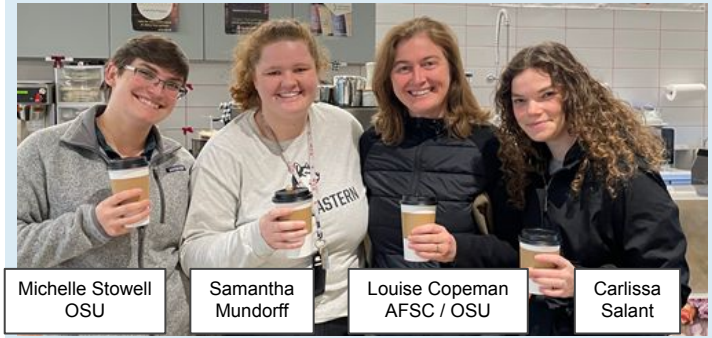
Juvenile Collection & Experimental Data


Ben Laurel, AFSC


Mary Beth Rew Hicks, AFSC


Paul Iseri, AFSC

Lipid Data


Michelle Stowell OSU

Samantha Mundorff

Louise Copeman AFSC / OSU

Carlissa Salant

Cod collaborators, contributors & funders

NOAA FISHERIES

SCHOOL OF AQUATIC & FISHERY SCIENCES

PACIFIC STATES MARINE FISHERIES COMMISSION



'Omics Data & Bioinformatics guidance


Ingrid Spies, AFSC


Sara Schaal, AFSC & UO


Wes Larson, AFSC

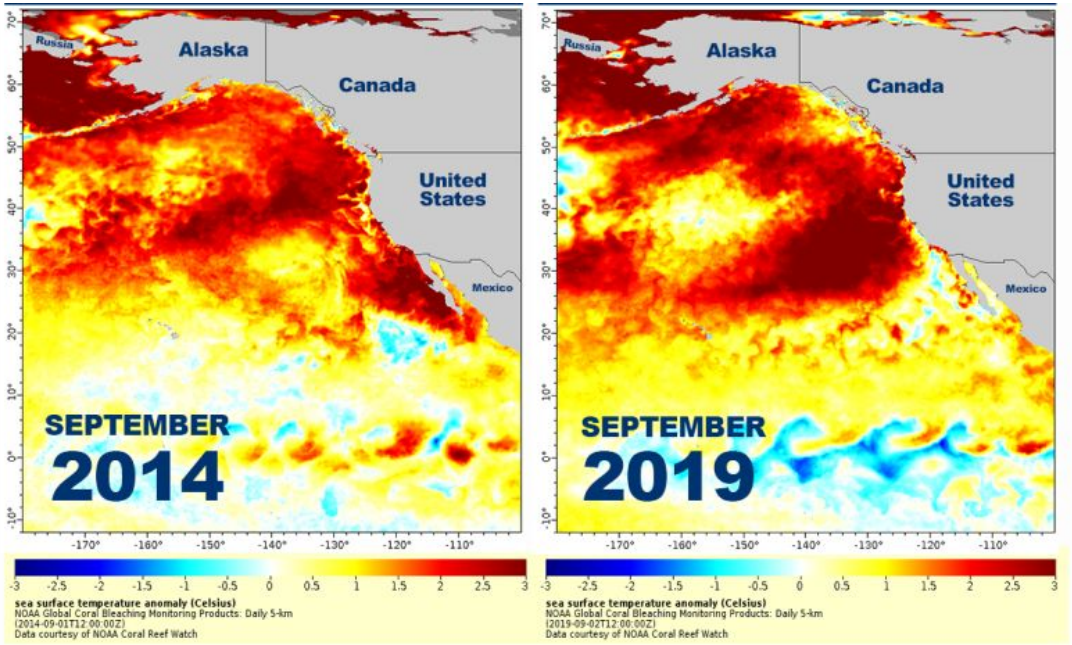

Steven Roberts, UW


Laura Timm, AFSC & UAF


Kathleen Durkin, UW

Study: Response of juvenile Pacific cod to temperature and genetic association

Heatwaves!



[NOAA feature story](), *September 05, 2019*



Pacific cod, *Gadus macrocephalus*

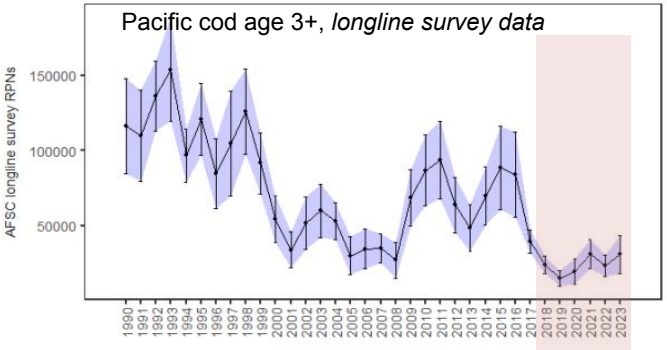
Extremely low cod numbers lead feds to close the Gulf of Alaska fishery for the first time

Alaska Public Media
Published December 6, 2019 at 8:22 PM AKST



Pacific cod (Photo Courtesy of Holland Dotts & the Alaska Marine Conservation Council)

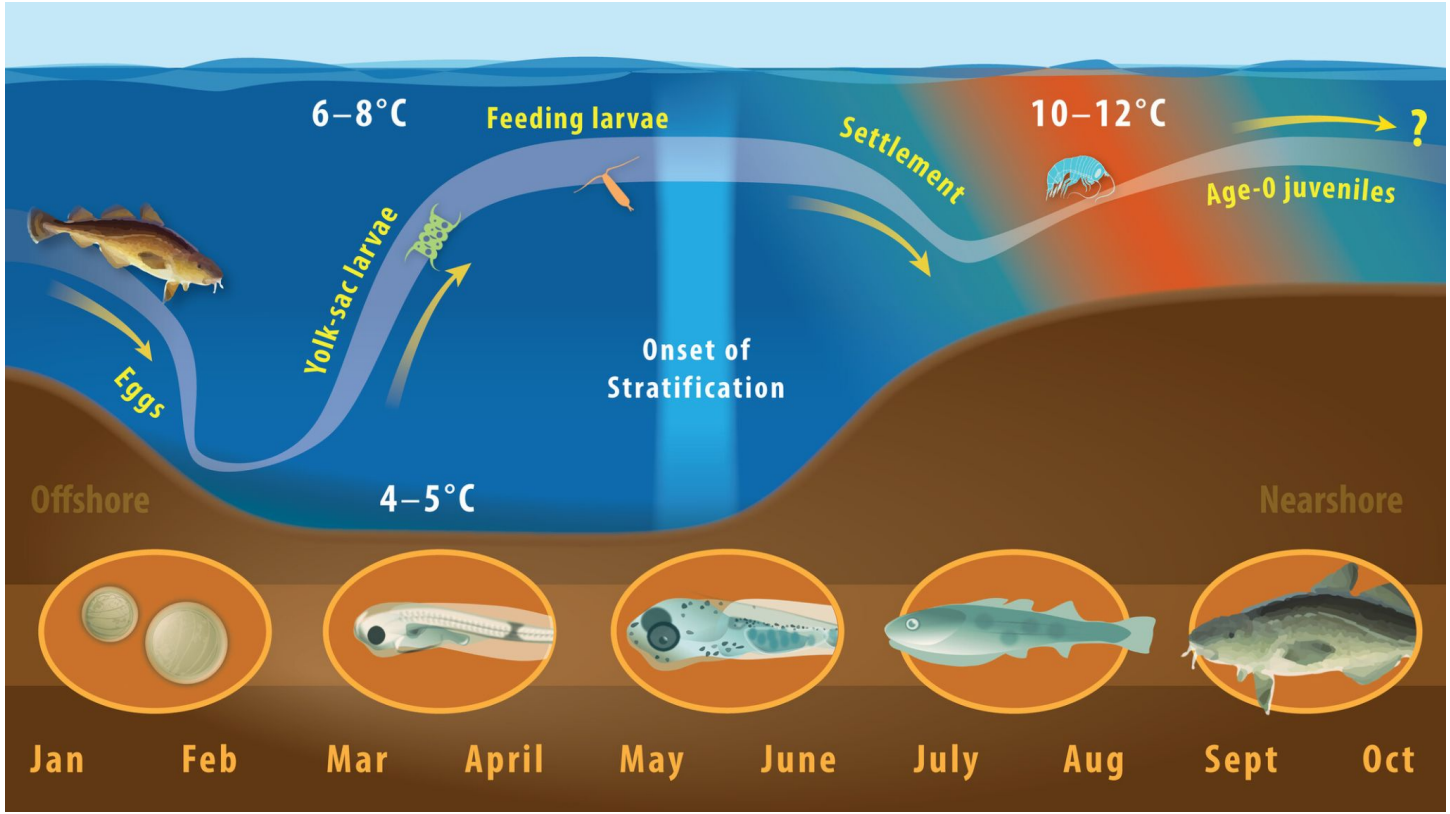
[Alaska Public Media](), *December 06, 2019*



[Stock Assessment Report, 2023, Gulf of Alaska](),



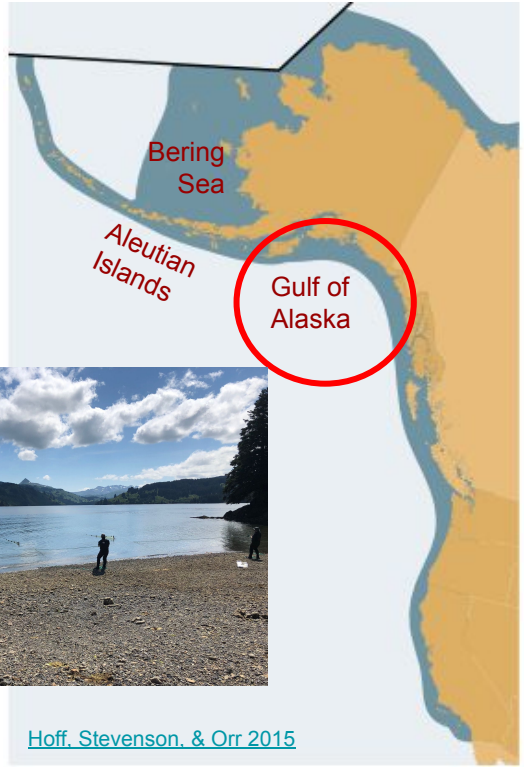
Pacific cod early life stages are very sensitive to temperature



Laurel et al. 2023, *in Fish & Fisheries*



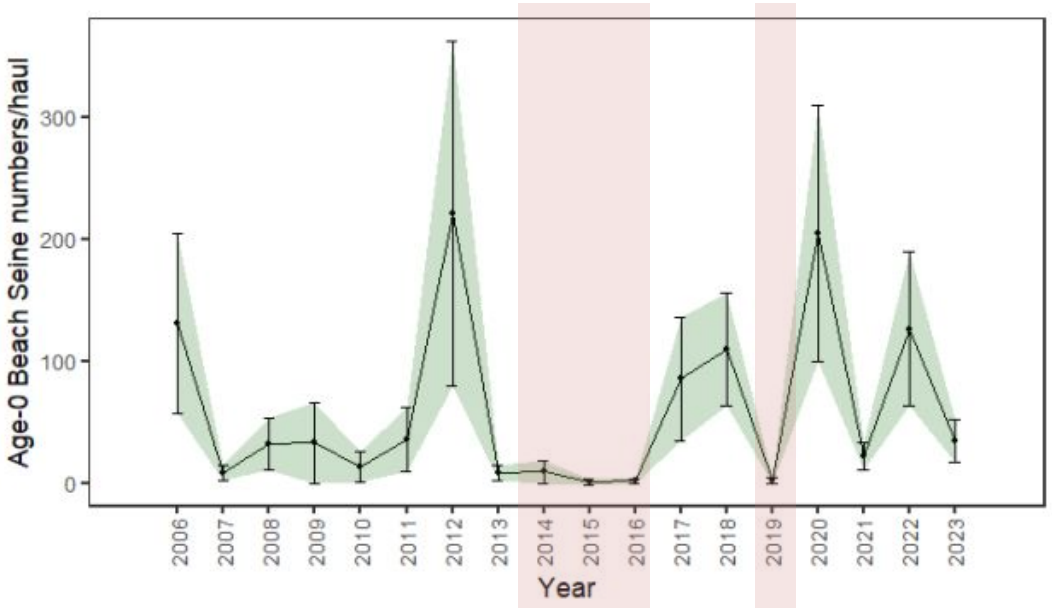
Low abundances likely related to low recruitment during heat waves





Hoff, Stevenson, & Orr 2015

Pacific cod juvenile (age-0) *beach seine survey CPUE*



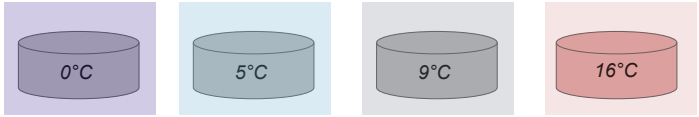
Stock Assessment Report, 2023, Gulf of Alaska,



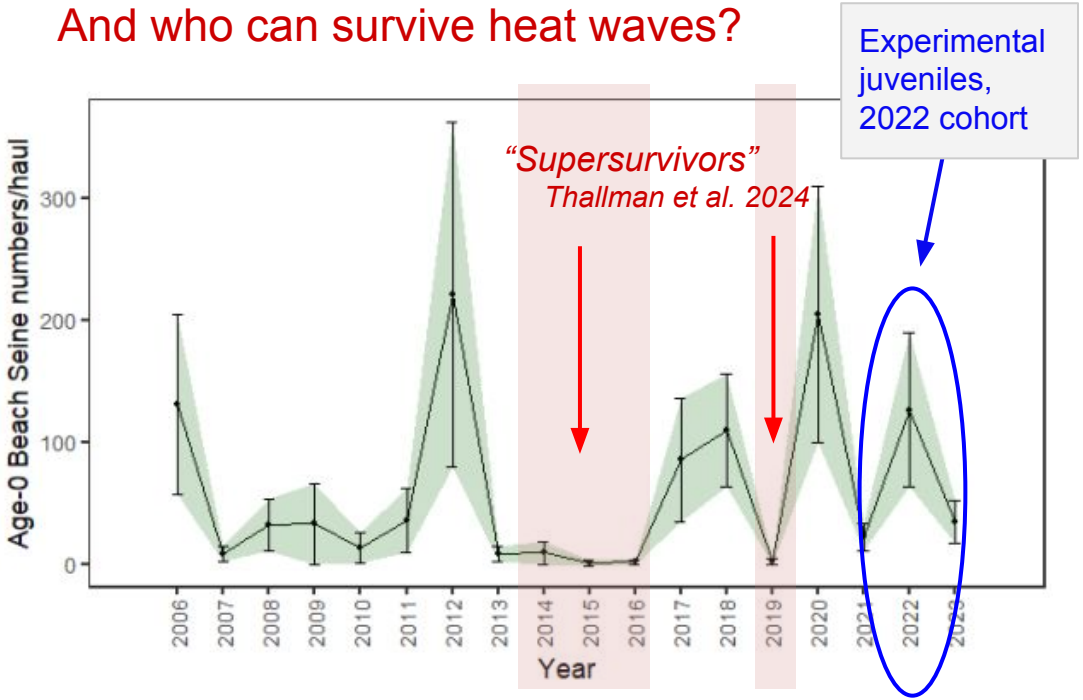
Question: How does temperature affect juvenile Pacific cod?

Juvenile Temperature Experiment





And who can survive heat waves?



Stock Assessment Report, 2023, Gulf of Alaska,



“Genome to Phenome” dataset across temperatures for Pacific cod





→ Individuals were tagged ←

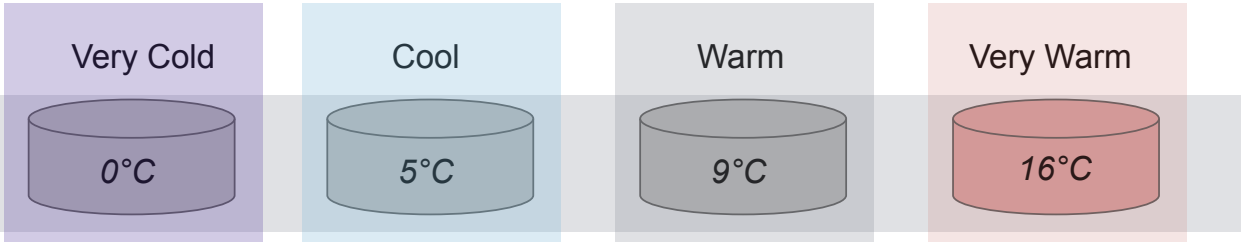
a. Genetics (lcWGS)

b. Gene expression in liver, n=18/temp (RNASeq)

c. Fitness Traits
   - Growth rates
   - Liver & Body Condition
   - Liver lipid concentrations (n=25/temp)
   - Survival

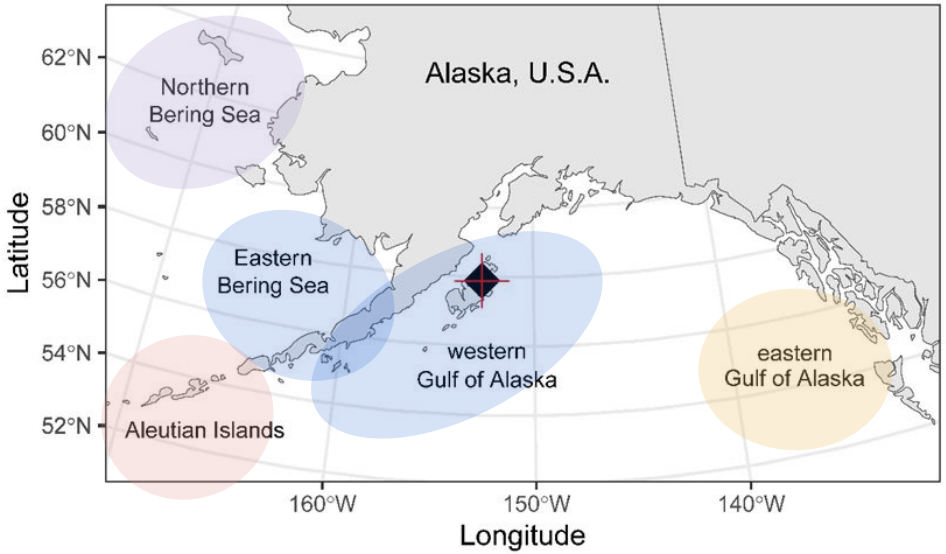


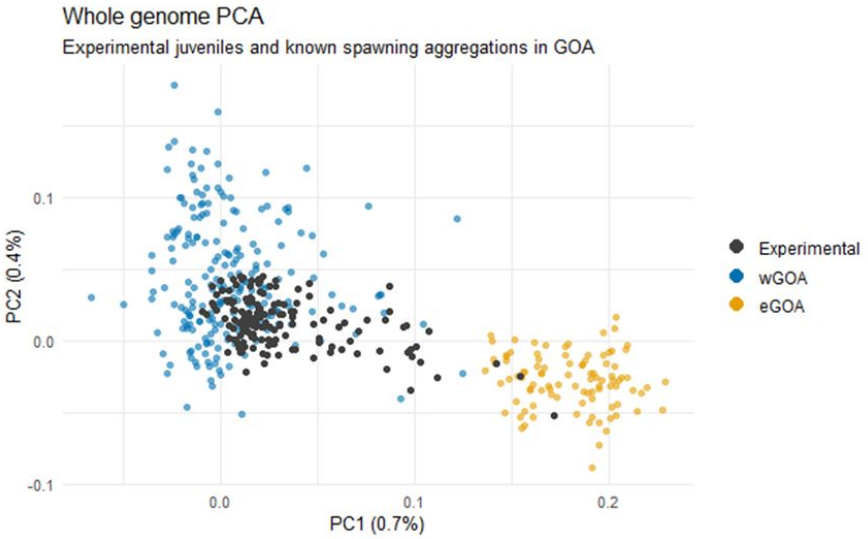
n=40 / temp.
4 week treatment



Genetics was used to identify population represented in experiment.

Most juveniles from west/central Gulf of Alaska, some with eastern GOA ancestry



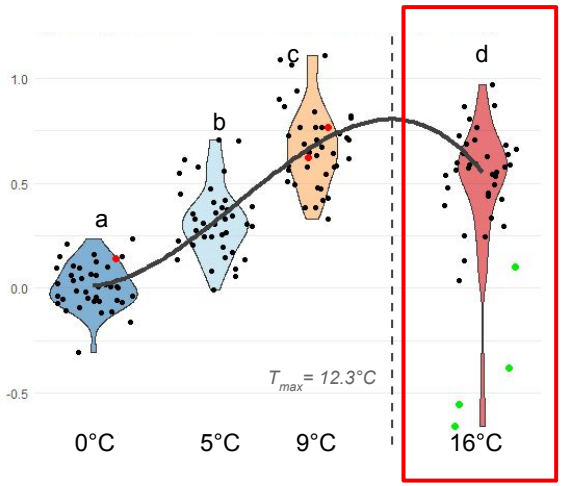


Schaal et al. 2026, https://doi.org/10.1111/eva.70174

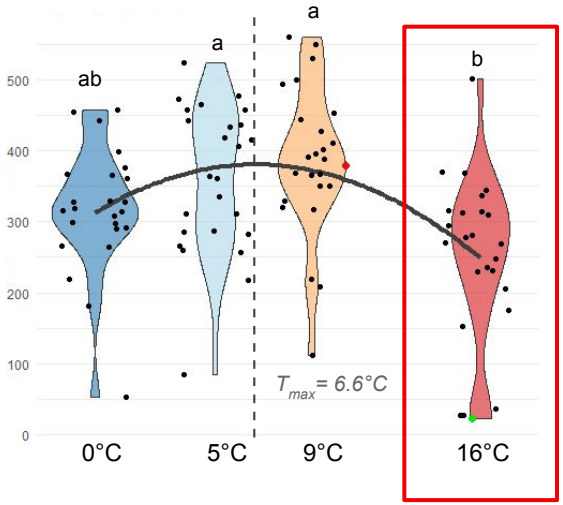


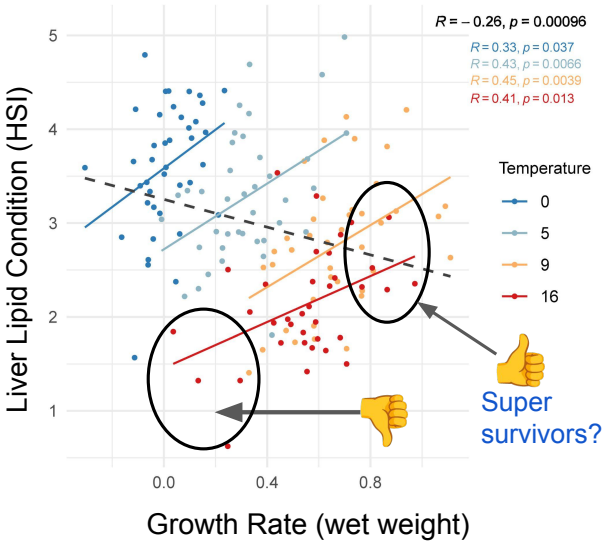
Warming slows growth, reduces energy reserves (lipid), & increases mortality - but growth & lipids are positively correlated

Growth Rate (weight, %/day)



Lipid Content in liver (mg)



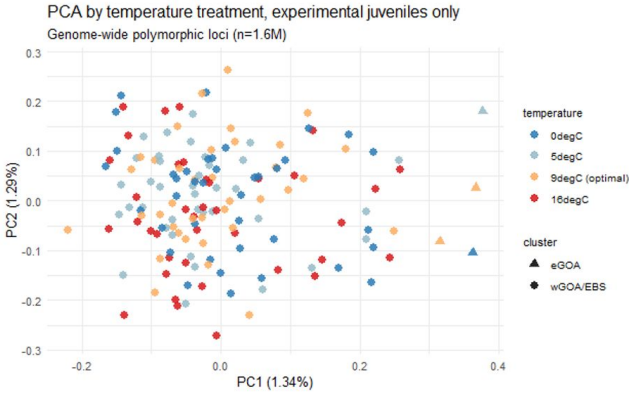


n=40/temperature

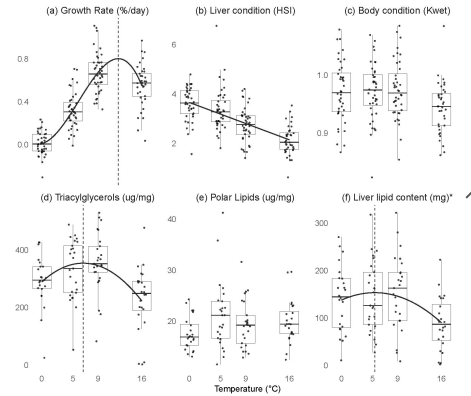
In press, GCB-Comm



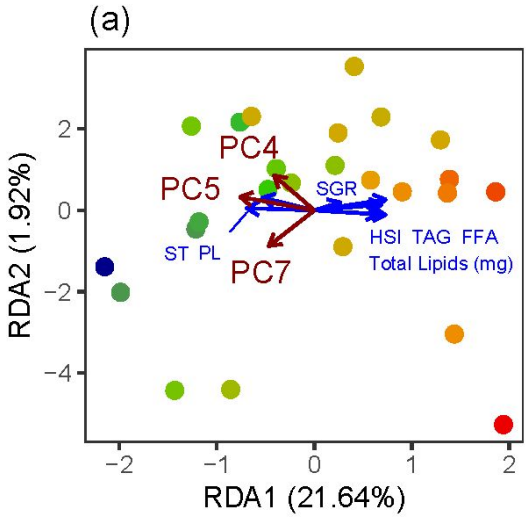
Associations among phenotypes, genetic variation in juveniles exposed to high temperature (16°C) explored using RDA

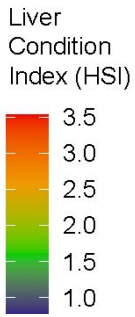


lcWGS PCs

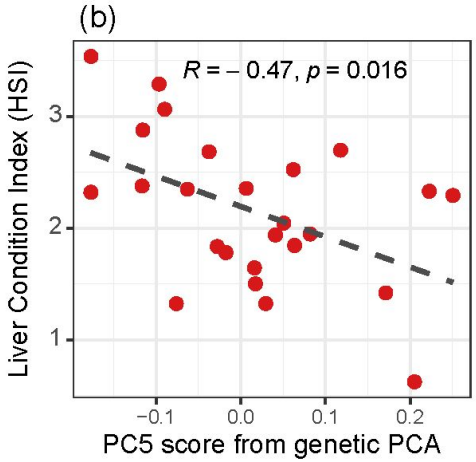


Phenotypes





~25% of phenotypic variation explained by RDA model with three PCs capturing ~4% of genetic variation in lcWGS



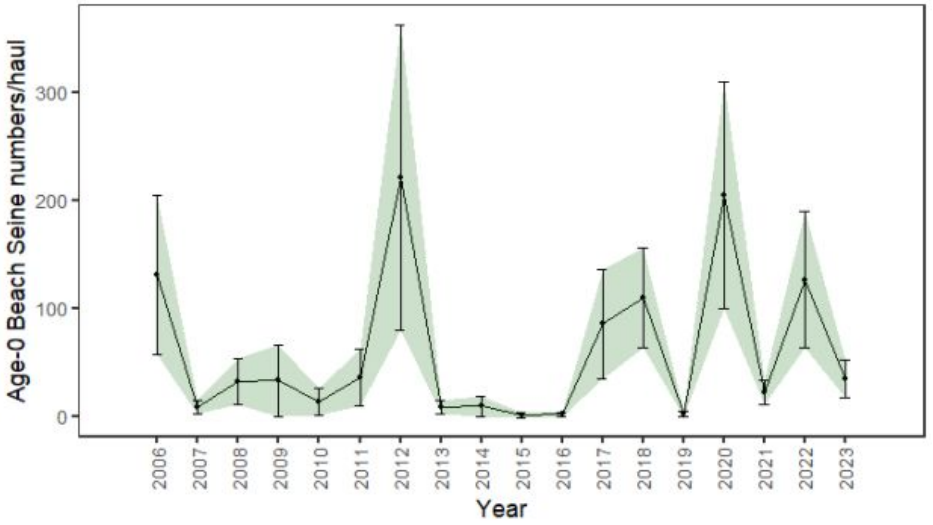
Study: Pacific cod spawn timing variation and juvenile genetic time series

Pacific juvenile genetic data from AFSC Kodiak beach seine survey



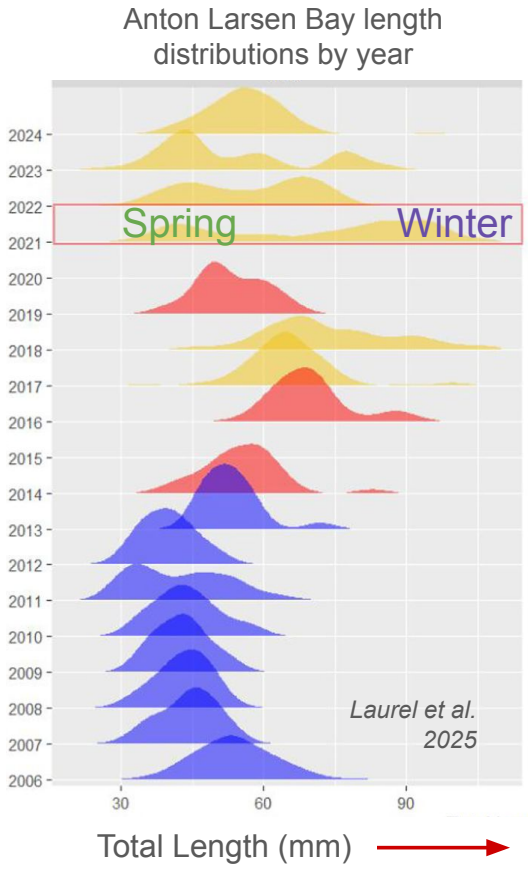
Hoff, Stevenson, & Orr 2015

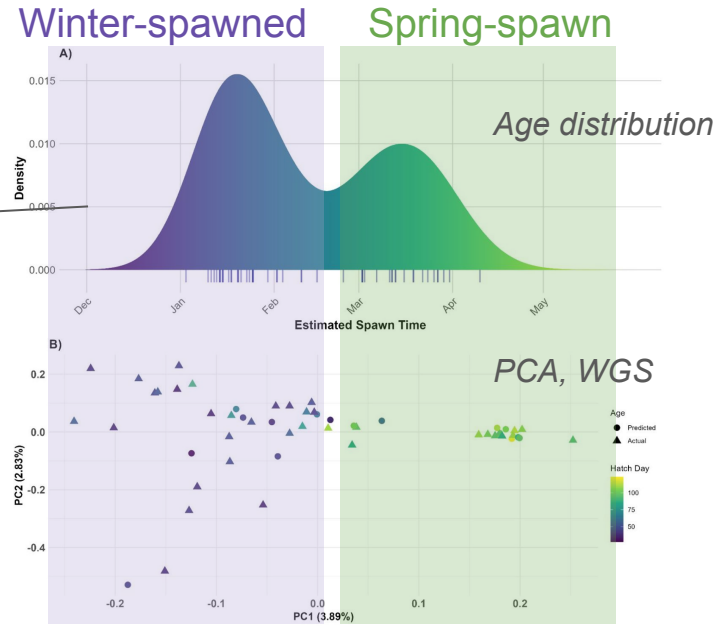
Pacific cod juvenile (age-0) *beach seine survey CPUE*



Stock Assessment Report, 2023, Gulf of Alaska,

Genetically-linked spawn/hatch timing variation in GOA Pacific cod

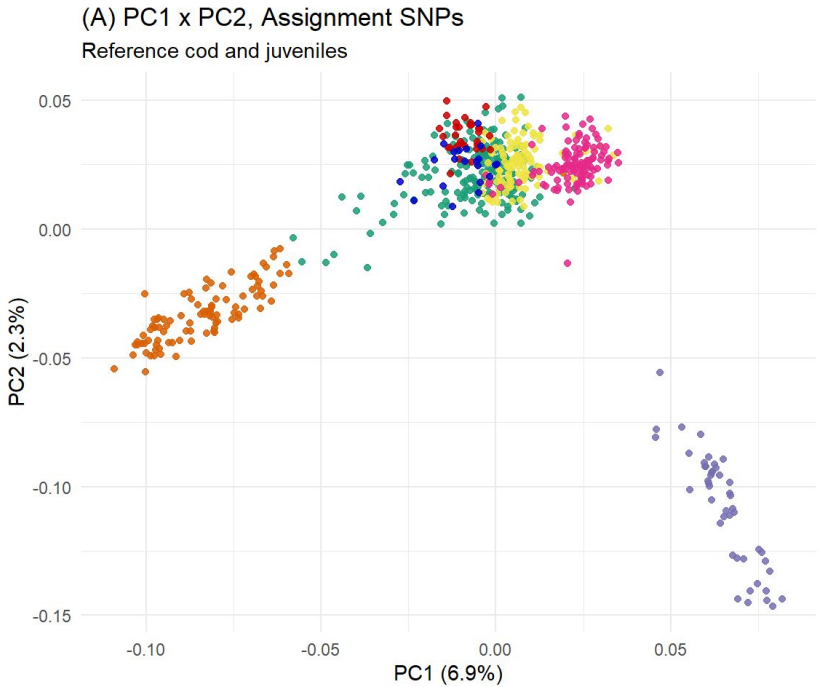


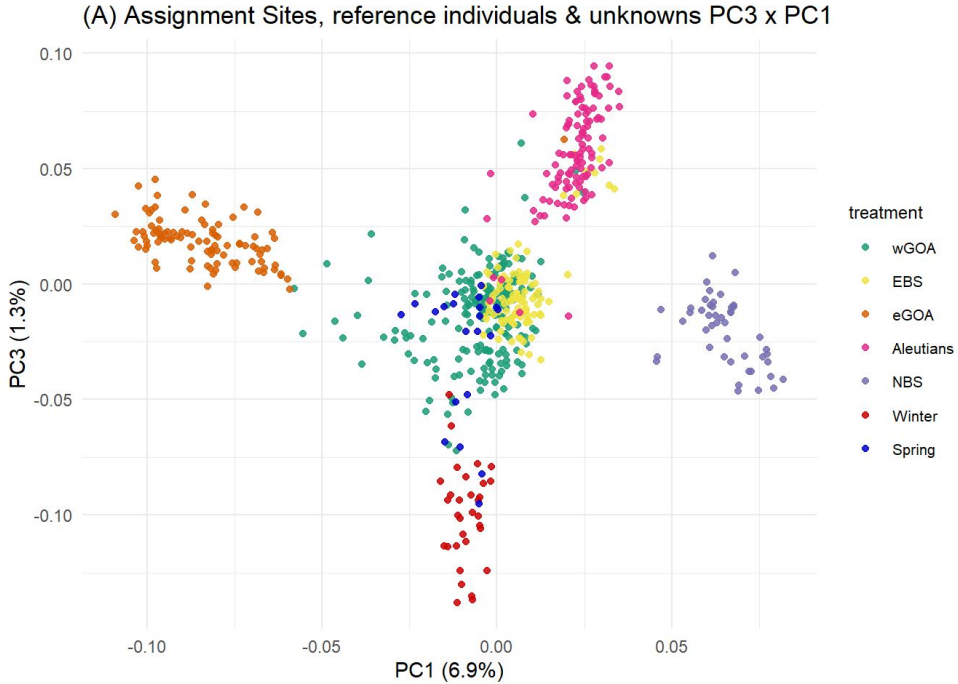


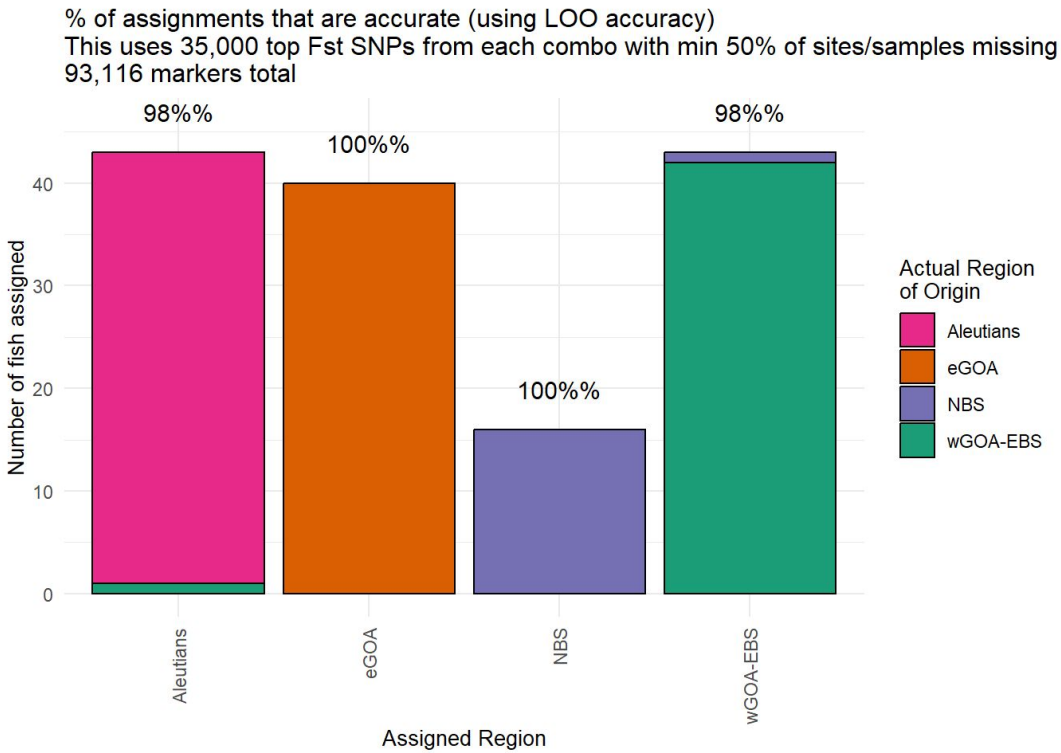
First evidence of genetically distinct winter- and spring-spawning components of Pacific cod in the Gulf of Alaska

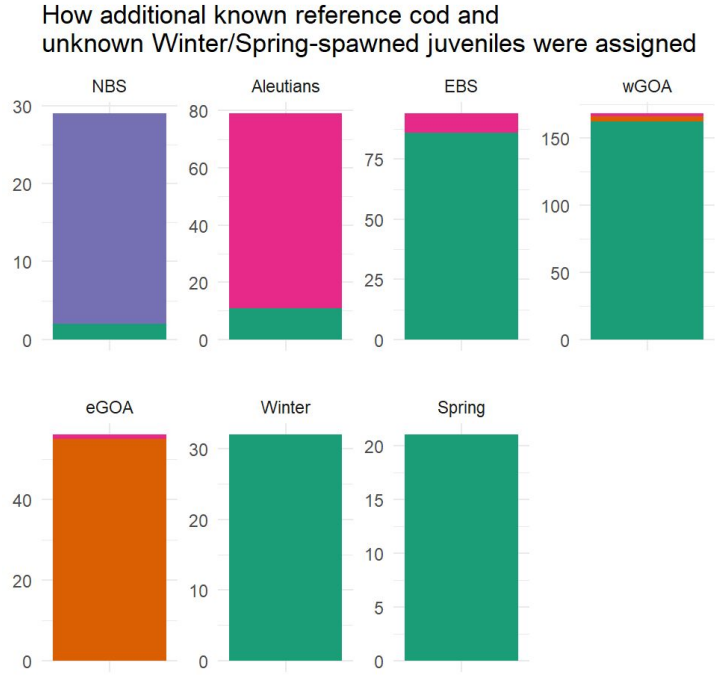
Authors: Benjamin Laurel, Jessica Miller, Louise Copeman, Alisa Abookire, Laura Spencer, Sara Schaal, Mary Beth Rew Hicks, Nicholas Strait, Lauren Rogers, and Ingrid Spies | AUTHORS INFO & AFFILIATIONS

Publication: Canadian Journal of Fisheries and Aquatic Sciences • 16 January 2026 • https://doi-org.offcampus.lib.washington.edu/10.1139/cjfas-2025-0168


(A) PC1 x PC2, Assignment SNPs
Reference cod and juveniles
0.05
0.00
-0.05
-0.10
-0.15
PC2 (2.3%)
-0.10
-0.05
0.00
0.05
PC1 (6.9%)


(A) Assignment Sites, reference individuals & unknowns PC3 x PC1
0.10
0.05
0.00
-0.05
-0.10
PC3 (1.3%)
-0.10
-0.05
0.00
0.05
PC1 (6.9%)
treatment
wGOA
EBS
eGOA
Aleutians
NBS
Winter
Spring


% of assignments that are accurate (using LOO accuracy)
This uses 35,000 top Fst SNPs from each combo with min 50% of sites/samples missing
93,116 markers total
98%%
100%%
100%%
98%%
Number of fish assigned
0
10
20
30
40
Aleutians
eGOA
NBS
wGOA-EBS
Assigned Region
Actual Region of Origin
Aleutians
eGOA
NBS
wGOA-EBS


How additional known reference cod and
unknown Winter/Spring-spawned juveniles were assigned
NBS
Aleutians
EBS
wGOA
eGOA
Winter
Spring

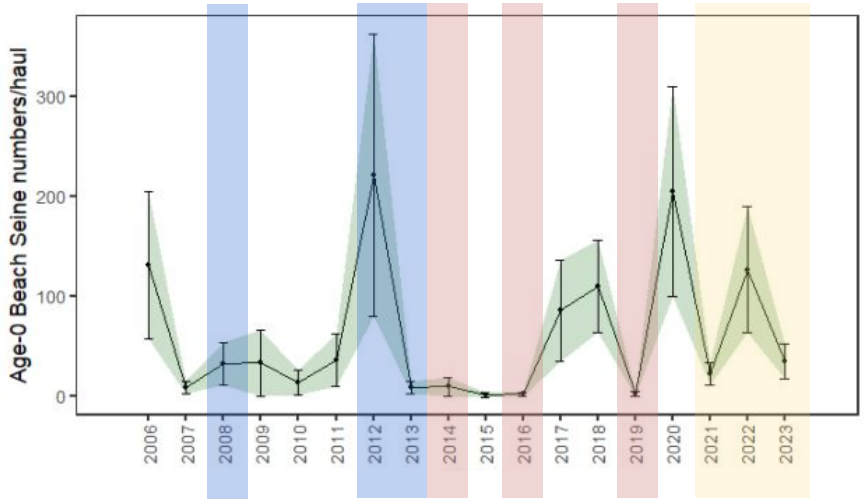






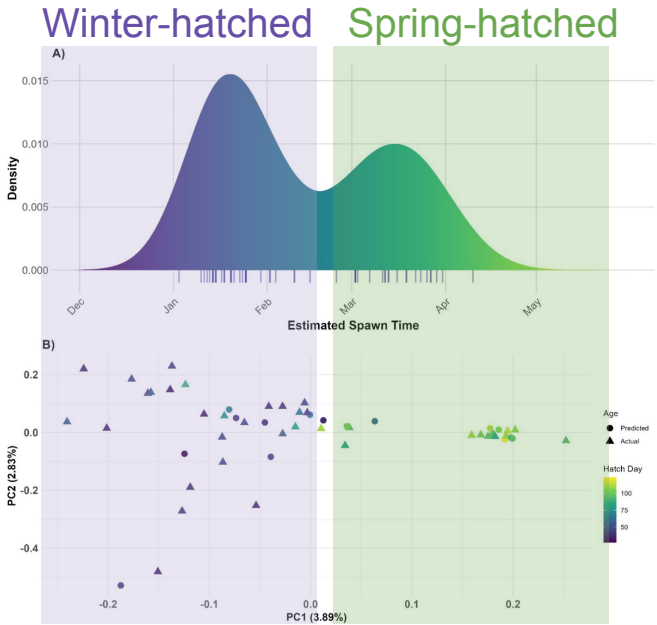
Juvenile genetic time series from Kodiak beach seine survey

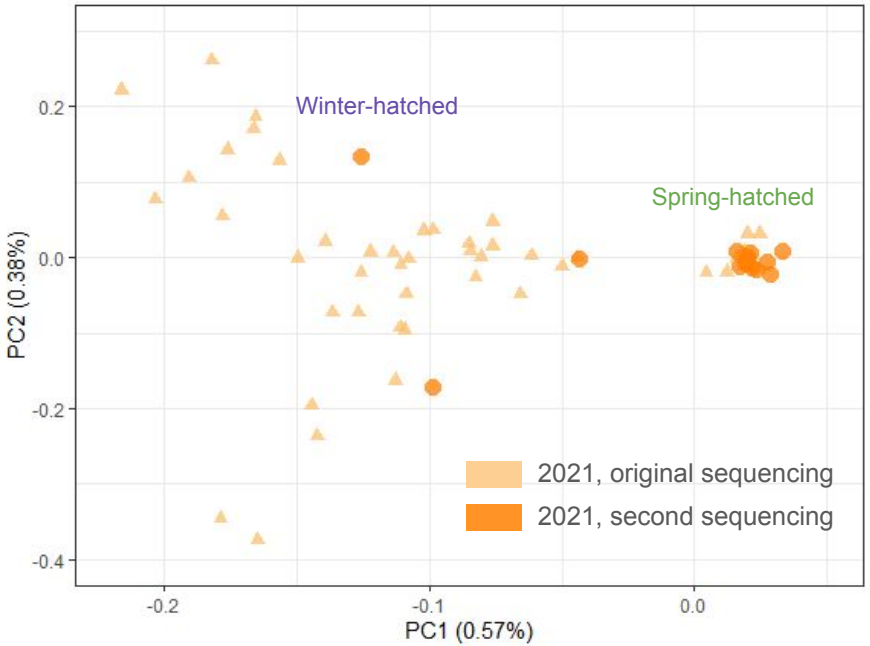
WGS from archived Pacific cod juvenile (age-0)



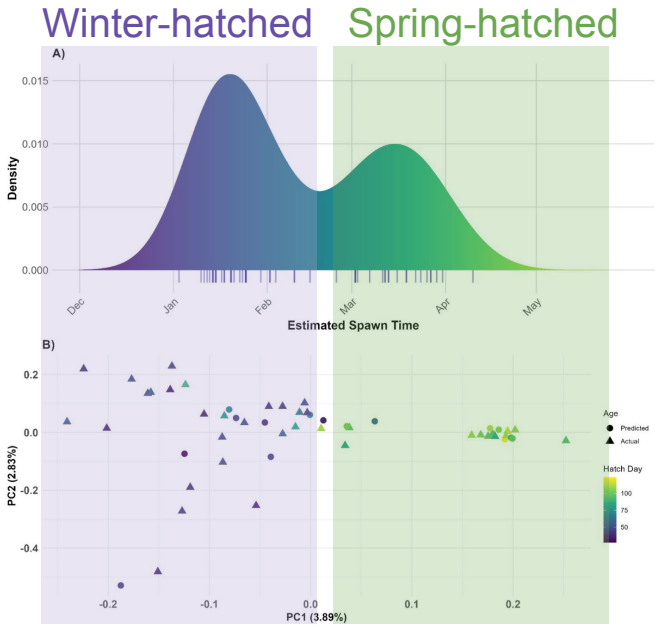
- Juveniles from **nine distinct years**, spanning 2008-2023
- 40+ per year*, all in July from the same location, Anton Larsen bay
- Medium coverage WGS (mean ~13x)
- Early stages of exploring data:
  - Who is there?
  - Changes over time

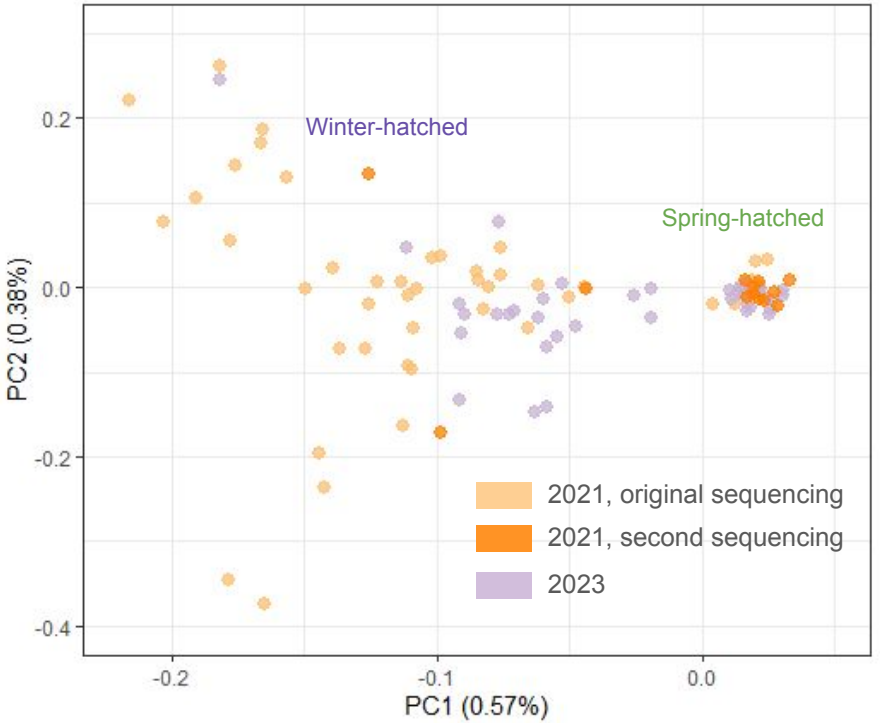
More support for genetically-linked spawn timing variation in GOA Pacific cod



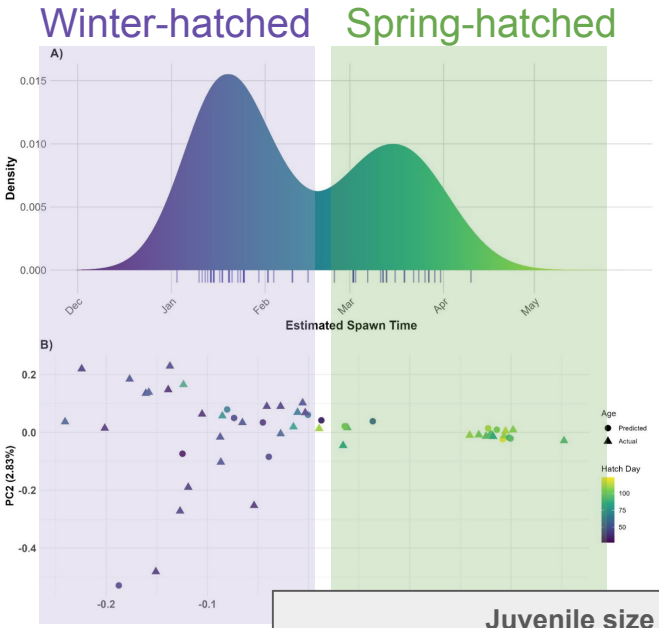


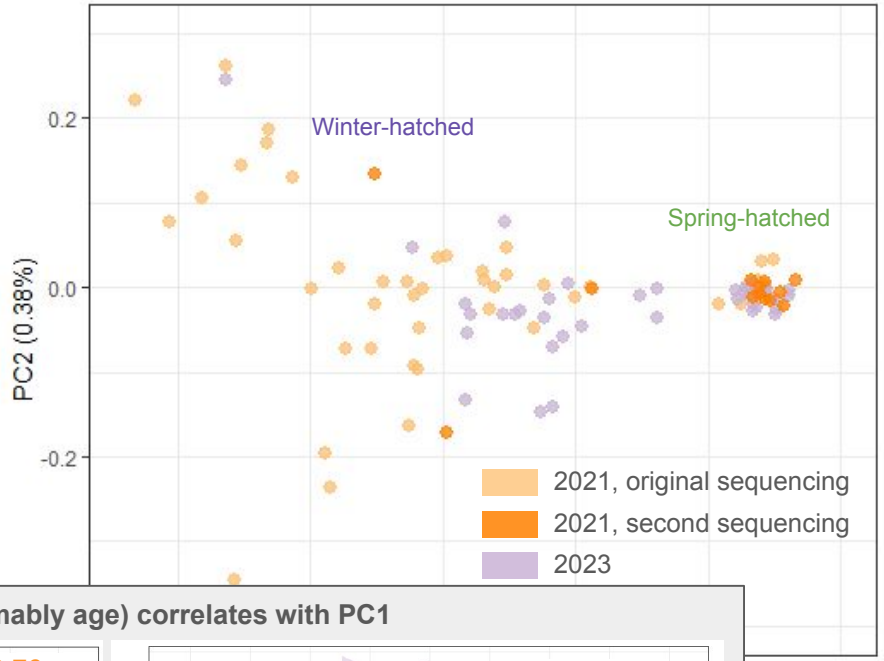
More support for genetically-linked spawn timing variation in GOA Pacific cod

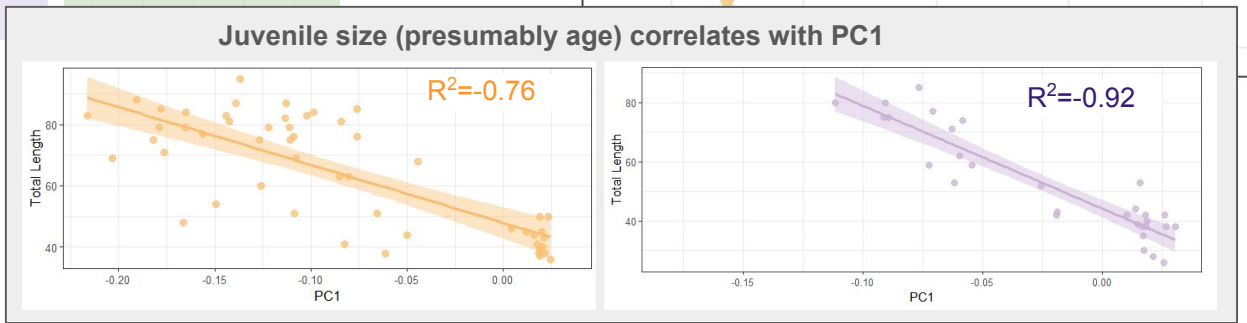




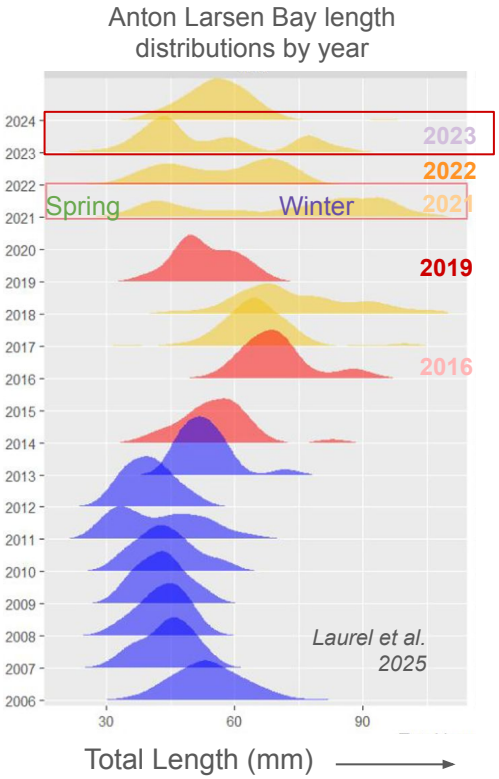
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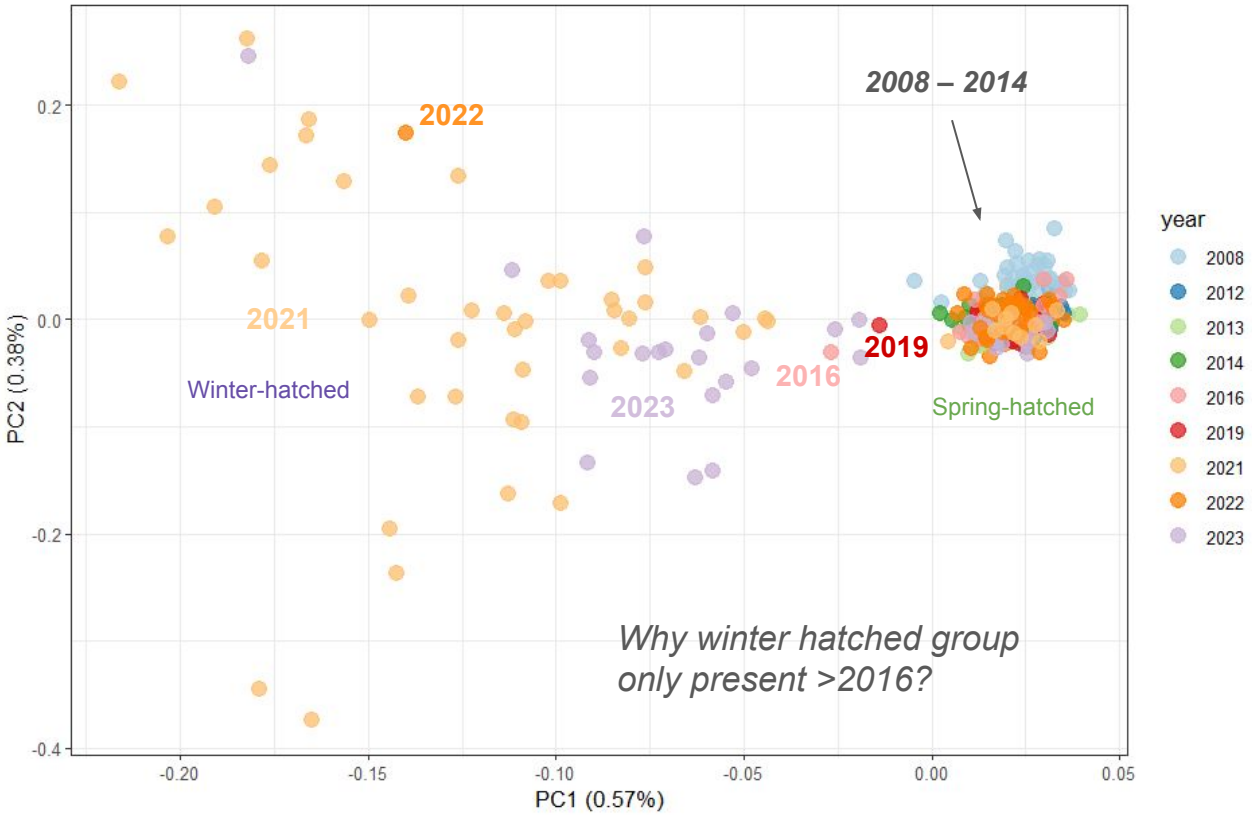




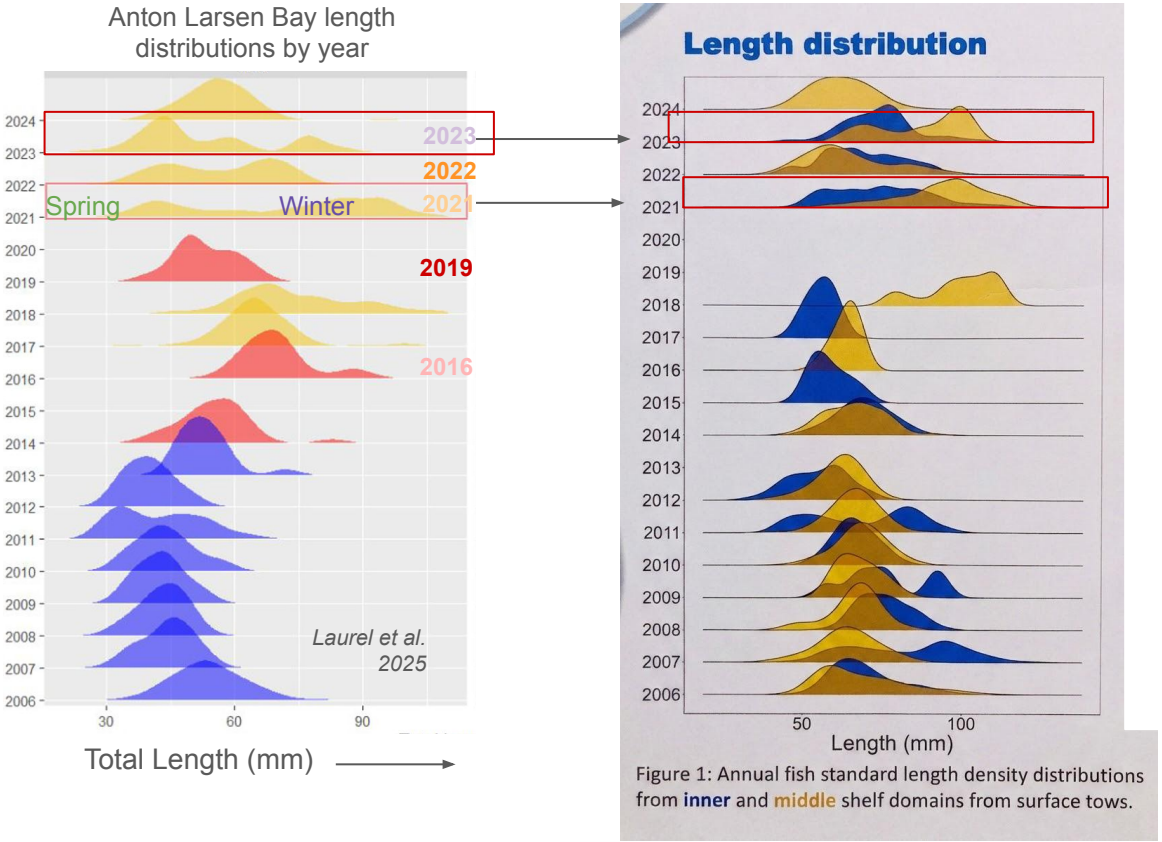


Most juveniles from time-series are like the "Spring-hatched" group, similar to Schaal et al. wGOA/EBS adult spawning reference data





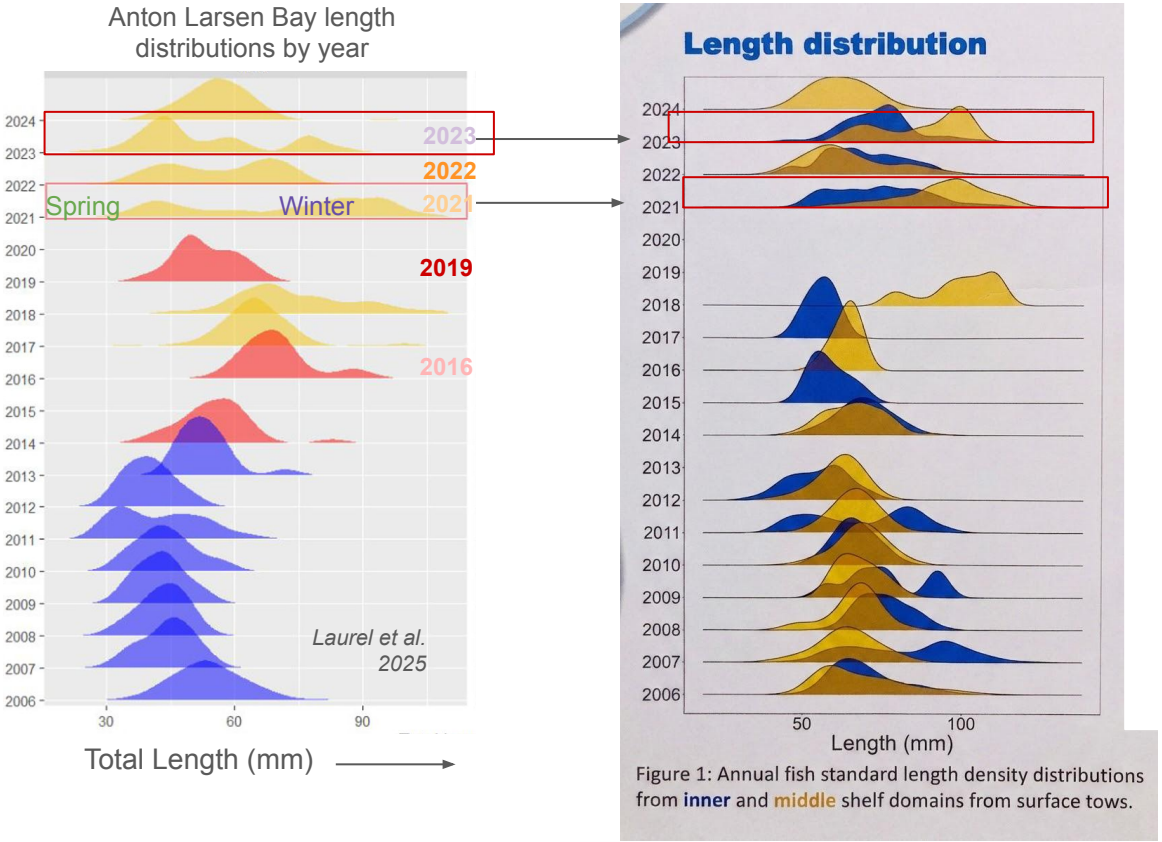
"Winter-hatched" group may be in Bering Sea, too.
Inner vs. middle shelf domain differences?



Eastern Bering Sea, surface & midwater tows

From Mariela Brooks' poster, AMSS 2026

Remaining questions



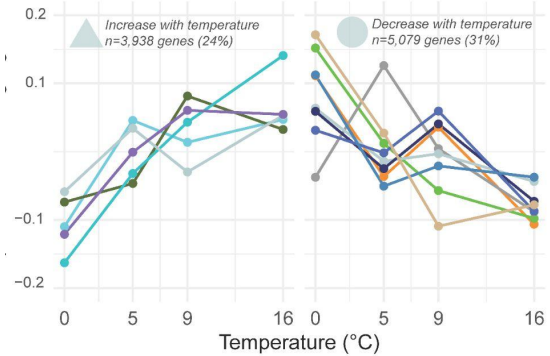
- Where does the“Winter” group come from?
- Why wasn’t it represented in the Schaal et al. spawning adult samples?
- Why is it more prevalent in recent years?
- Are they in the Bering Sea, too?
- Are they physiologically different?

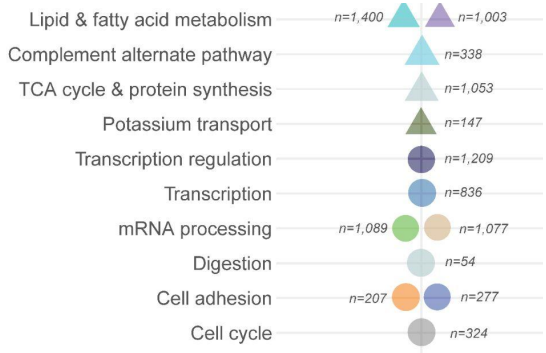
Study: Juvenile fitness-associated markers in Pacific cod – experimental results, and exploration through time in the field.

Are warm-tolerant alleles more prevalent since marine heatwaves?

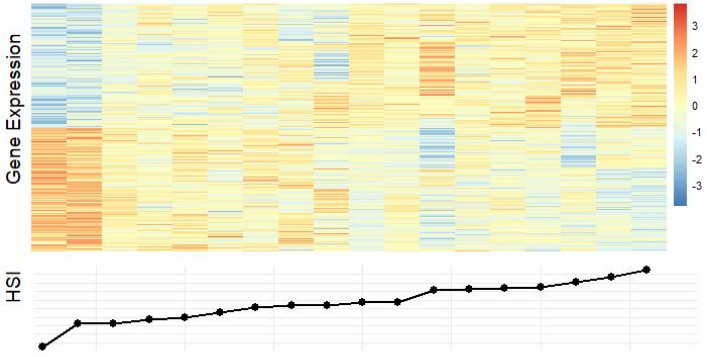
Genetic & expression markers associated with liver condition in warming

Liver expression across temperature



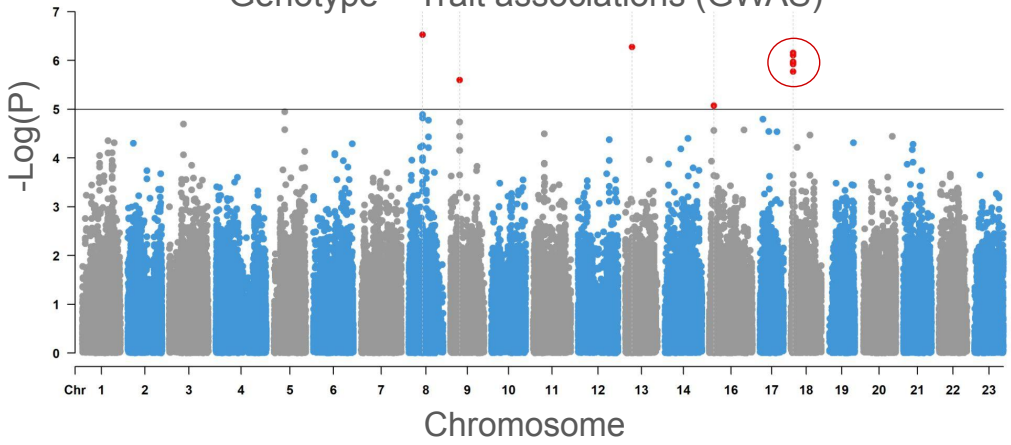


Liver expression ~ liver condition in warming

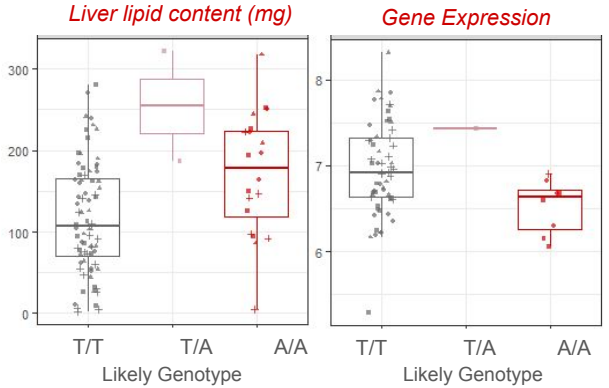


Blood expression + methylation, too!
NPRB proposal

Genotype ~ Trait associations (GWAS)

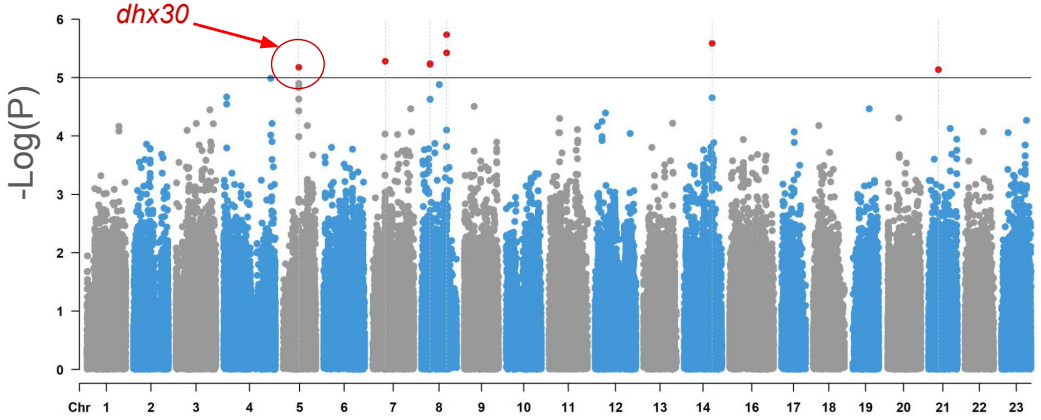


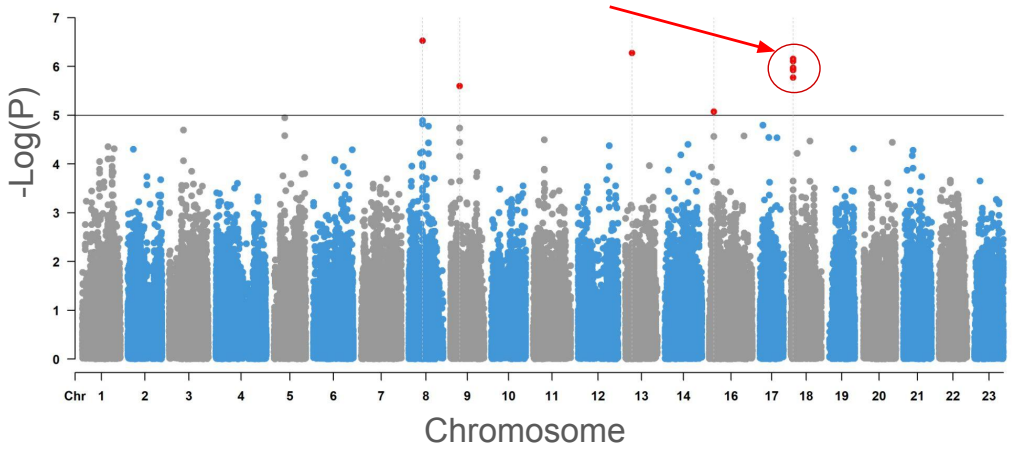
Expression ~ trait ~ genotype associations

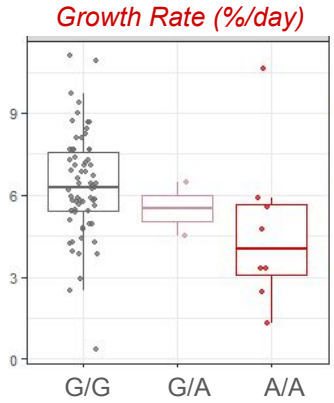


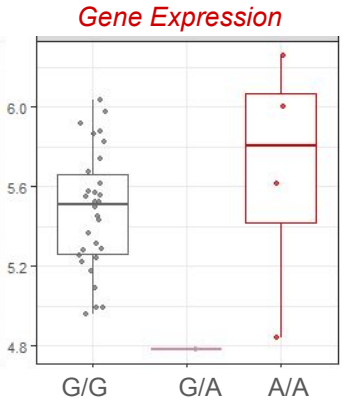


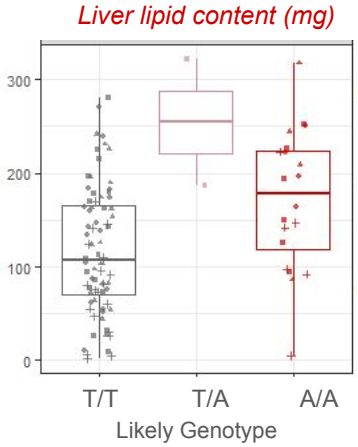
SNP markers putatively associated with **Growth Rate** by weight in Pacific cod juveniles

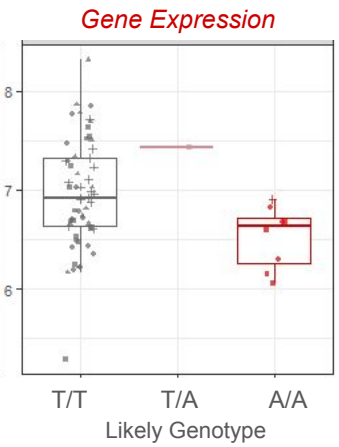














Polygenic scores, calculated from genotypes at markers associated with growth, liver lipid content, survival

POLYGENIC SCORE (PGS)

Combine many genetic effects into one score

1. GENOTYPE (DOSAGE)
0, 1, or 2 copies of the effect allele

SNP	You
SNP1	1
SNP2	2
SNP3	1
...	...
SNPm	0

×

2. EFFECT SIZE (β)
Effect of the allele from GWAS

SNP	β
SNP1	+0.12
SNP2	−0.08
SNP3	+0.05
...	...
SNPm	−0.03

=

3. POLYGENIC SCORE (PGS)
Sum across all SNPs

Your PGS
+0.23



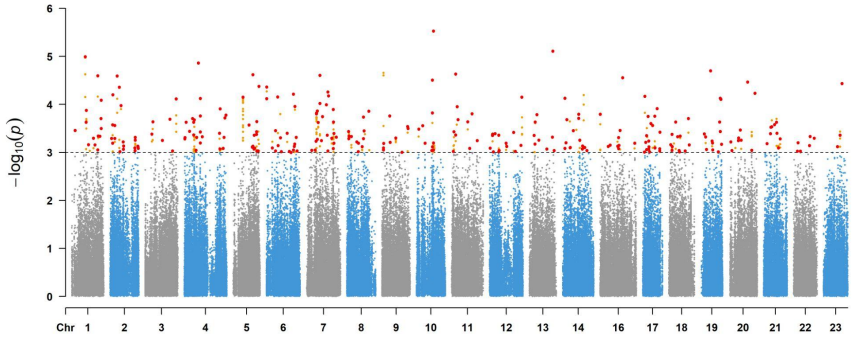
Lower PGS

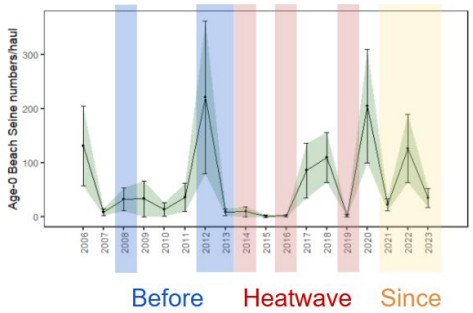
More "detrimental" alleles associated with ***lower growth, less lipid***

Higher PGS

More "beneficial" alleles associated with ***higher growth, more lipid***

Polygenic score for ***Liver Condition (HSI) uses 280 SNPs***





~40 each from 9 years, spanning 2008-2023

~20 in 2019, ~60 in 2021



Polygenic scores for HSI, growth rate (weight) increase over time in wild-caught juveniles (*preliminary!*)

Median Polygenic Scores ± IQR (q25-q75)

Larger livers
Faster growth

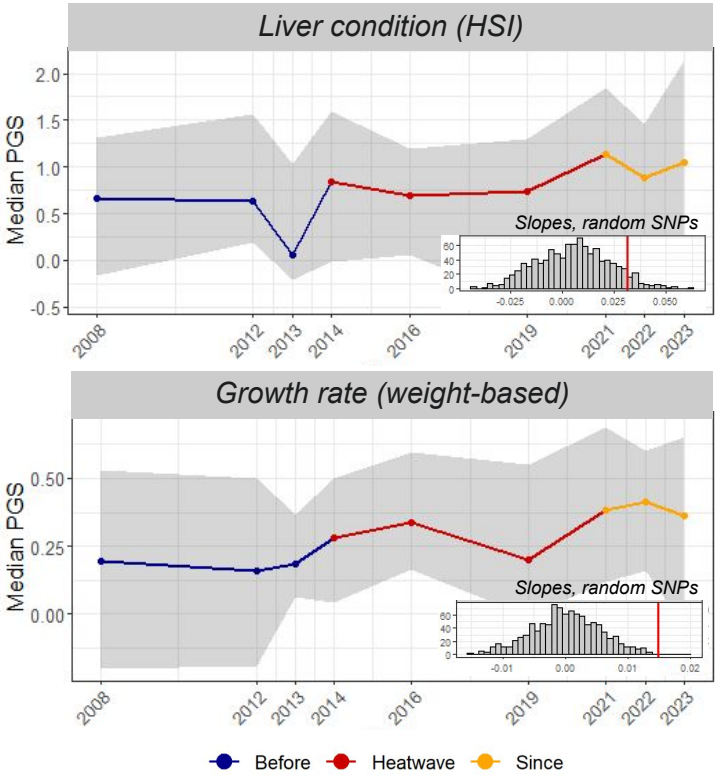
+

↑

↓

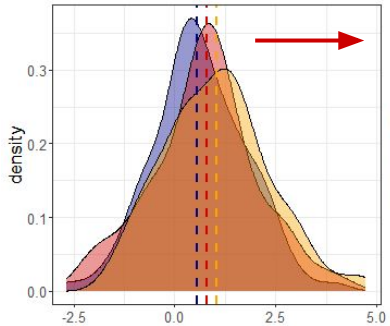
-

Smaller livers
Slower growth



Weak positive trend for HSI

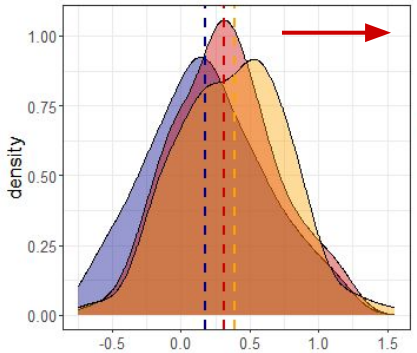
Slope: +0.028 / year
Total change: +0.42
p (LM): 0.068
R^2: 0.016



PGS for HSI

Positive trend for growth rate

Slope: +0.015 / year
Total change: +0.22
p (LM): 0.0006
R^2: 0.033



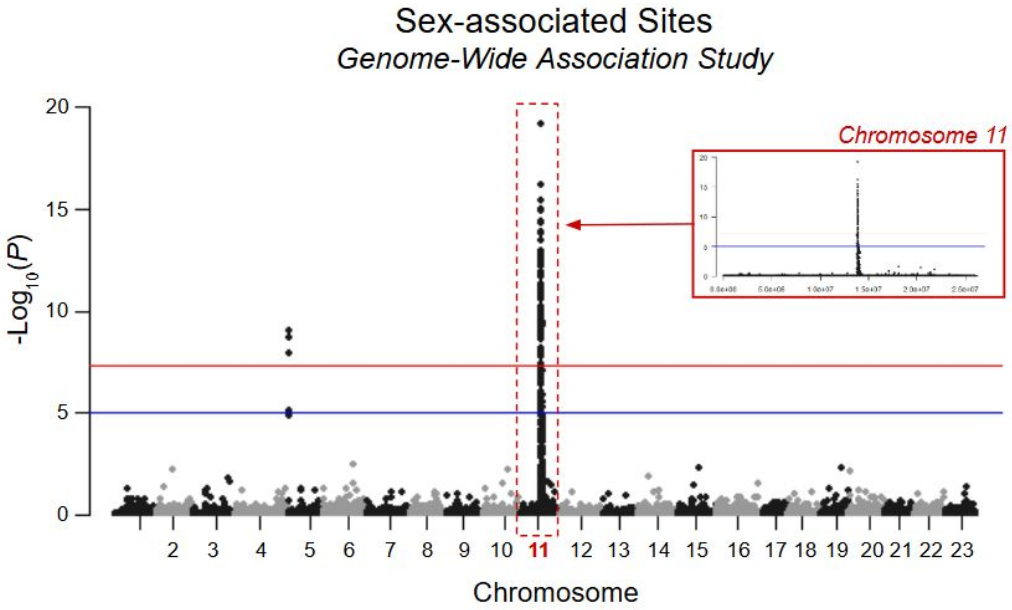
PGS for growth rate

Note, lots of within-year variability

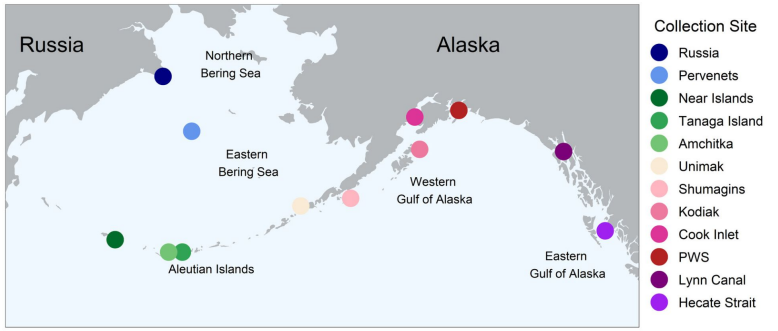
Study: Pacific cod sex markers

Pacific cod sex markers

103 females, 63 males

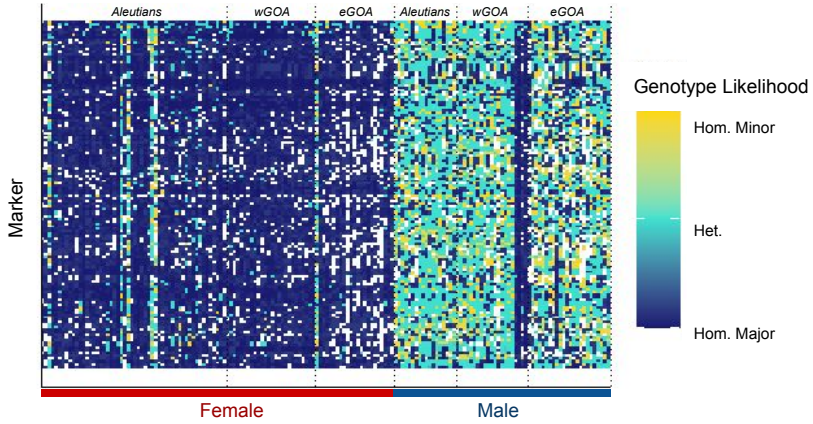


A

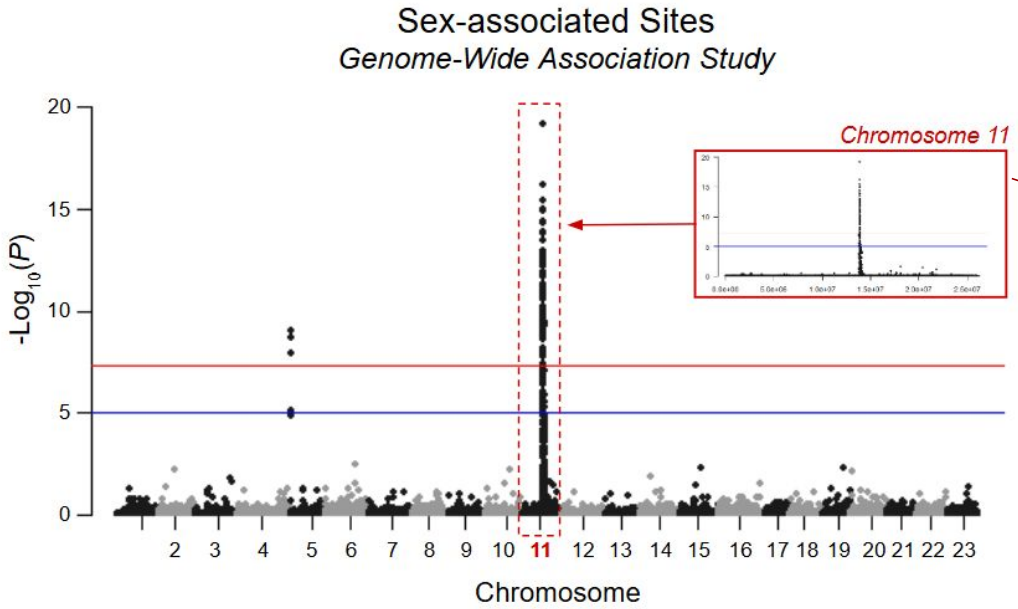


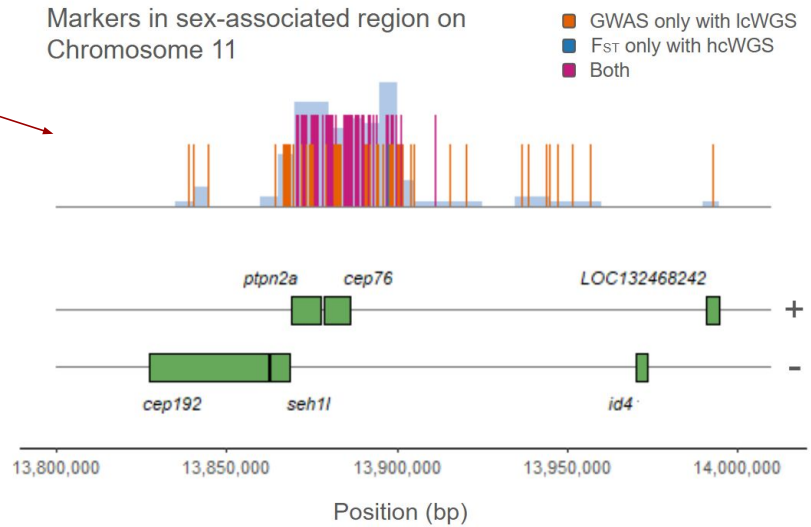
[Schaal et al. 2026]

Sex-associated markers on Chromosome 11

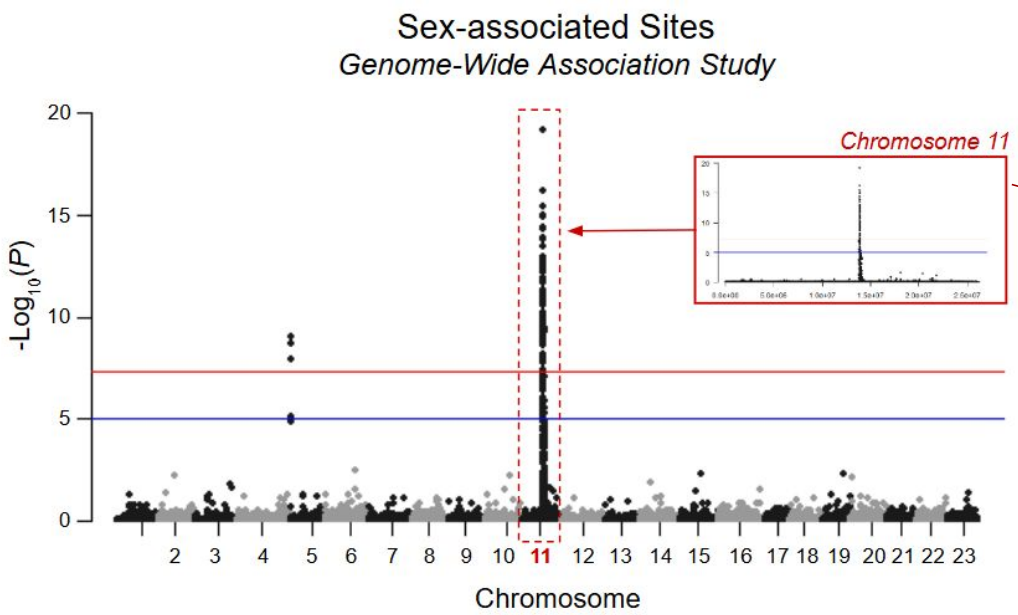


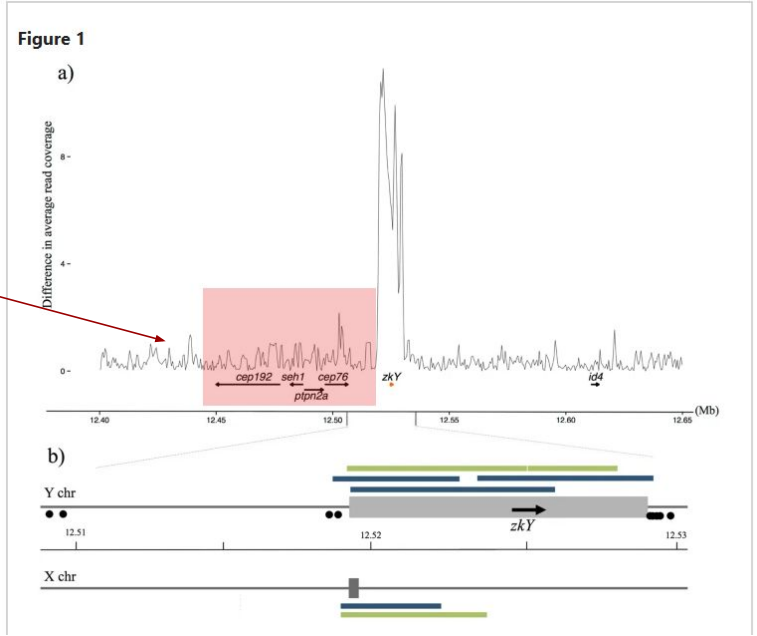
Pacific cod sex markers





Pacific cod sex markers





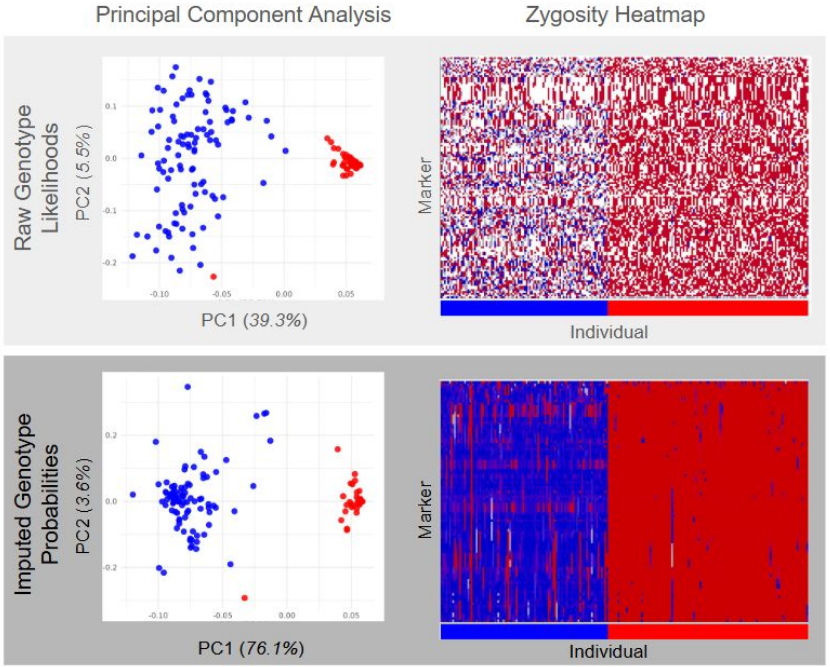
Article | Open access | Published: 15 January 2019

Characterization of a male specific region containing a candidate sex determining gene in Atlantic cod

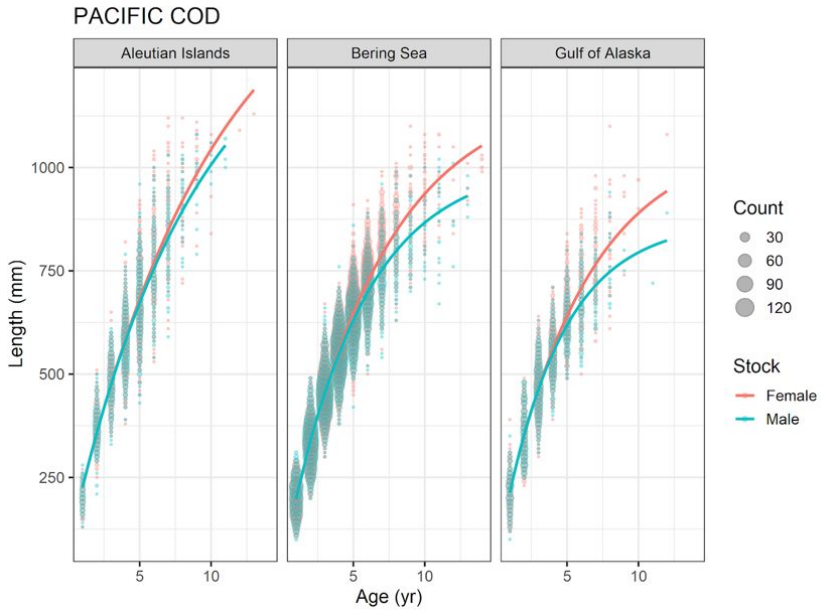
Tina Graceline Kirubakaran, Øivind Andersen, Maria Cristina De Rosa, Terese Andersstuen, Kristina Hallan, Matthew Peter Kent & Sigbjørn Lien

Scientific Reports **9**, Article number: 116 (2019) | Cite this article

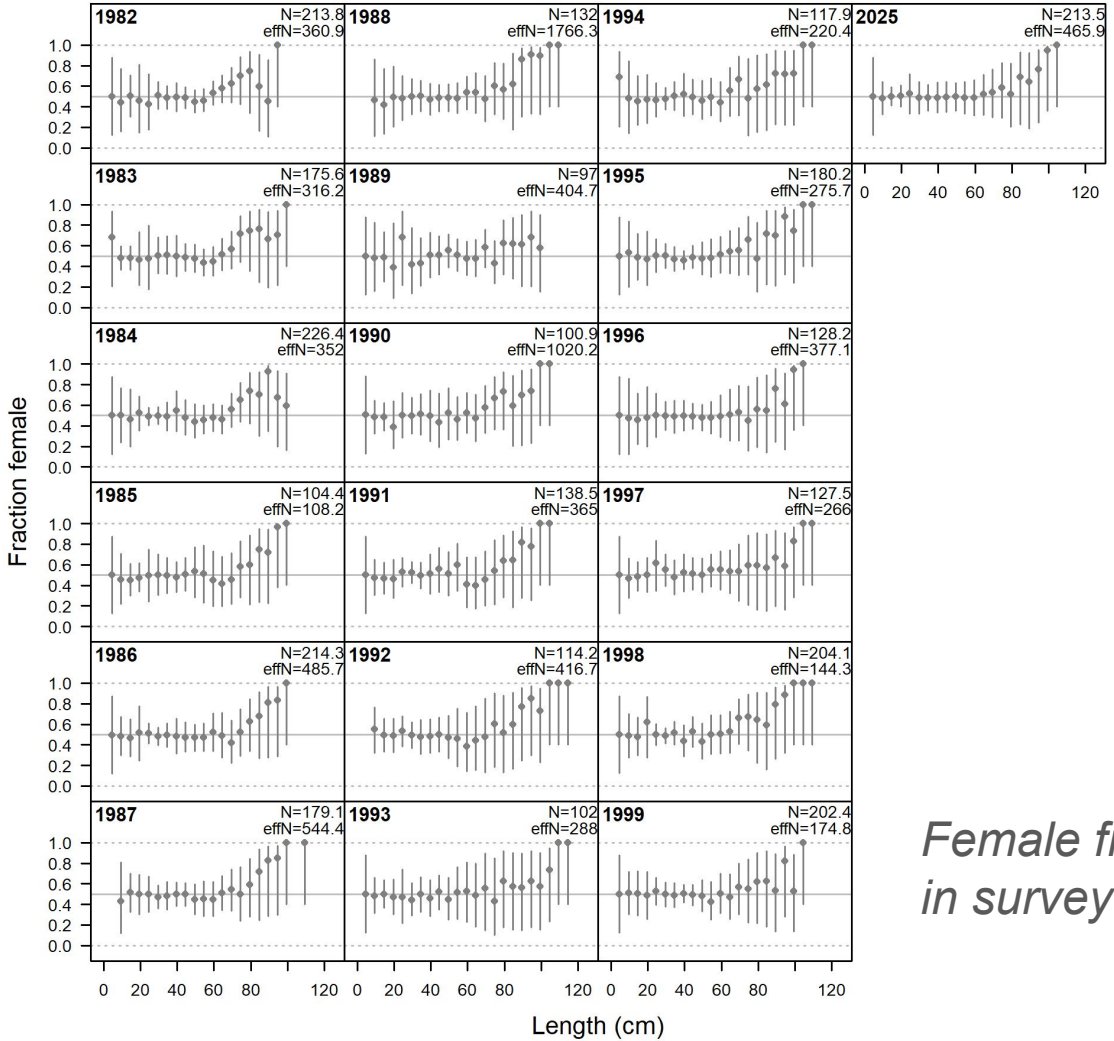
Predicting sex using lcWGS



- Sex assigned to tagged fish, early life stages
- Future integration with things (CKMR, epigenetic aging)
- **AFSC exploring split-sex model, size-at-age varies by sex,** sex ratios skewed



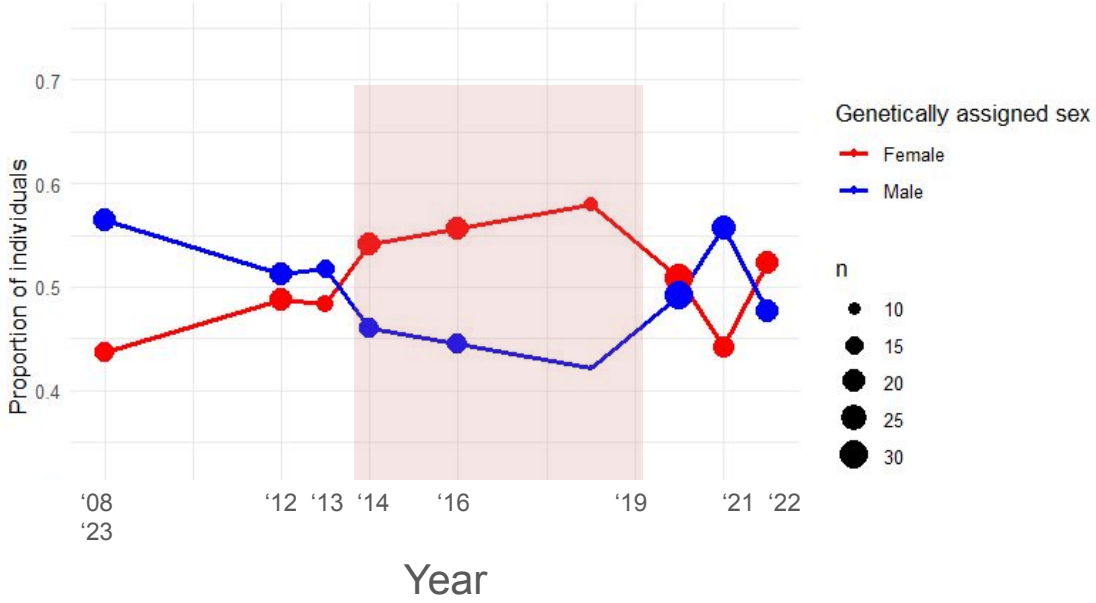
Highest accuracy –
Heterozygosity of imputed genotypes at sex markers
(219/220 matched phenotypic sex)



- Sex assigned to tagged fish, early life stages
- Future integration with CKMR, epigenetic aging
- **AFSC exploring split-sex model –** size-at-age varies by sex, **sex ratios skewed in fishery, survey**
  - ***Higher male natural mortality, or fishery mortality?***

Female fraction by length in survey

Juvenile sex ratio in Kodiak nursery over time – maybe more females during heat waves years? *Need more data*



- Sex assigned to tagged fish, early life stages
- Future integration with CKMR, epigenetic aging
- AFSC exploring split-sex model – size-at-age varies by sex, sex ratios skewed in fishery, survey
  - → **track sex ratios by cohort**

Remaining questions:

- **Y-specific insertion with zkY?**
- **Genetic sex assign tools: GTSeq, benchop assay**
- **Higher male natural mortality, or fishery mortality?**
- **Natural mortality vary by environment?**

Collaborations with NOAA experimental laboratories









NOAA Newport Lab,
Behavioral Ecology Program





NOAA Kodiak Lab,
Shellfish Assessment Program

Experimental Data & Funding



Chris Long, AFSC



Jennifer Gardner, AFSC



Steven Roberts, UW



Crab collaborators, contributors & funders

NOAA OCEAN ACIDIFICATION PROGRAM

SCHOOL OF AQUATIC & FISHERY SCIENCES

NOAA FISHERIES

CICOES

Bioinformatics & analyses guidance



Ingrid Spies, AFSC



Sara Schaal, AFSC & UO



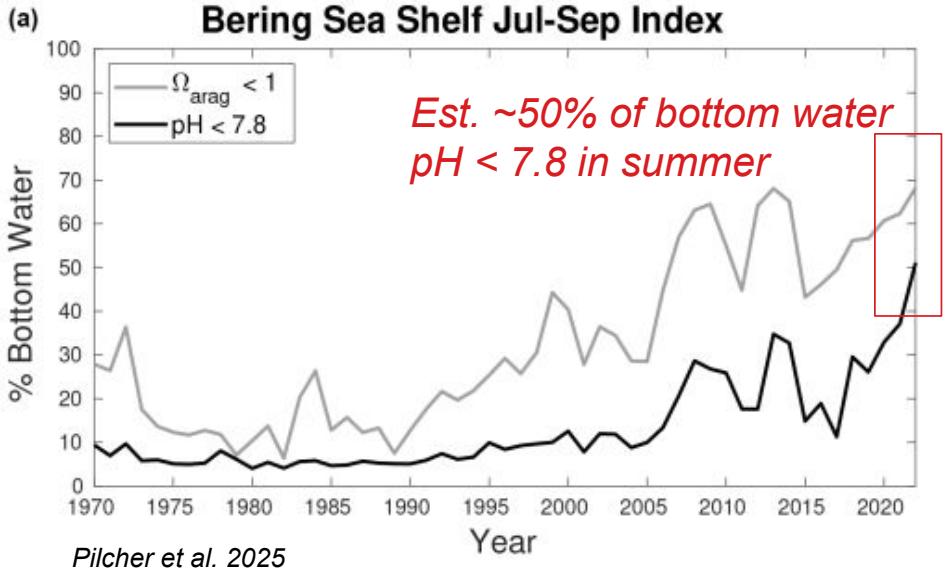
Natasha Howe , AFSC, Pacific States



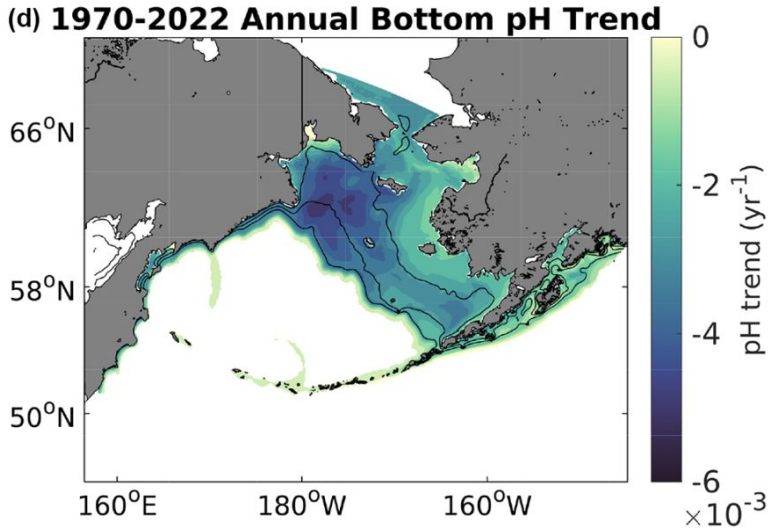
Laura Timm, AFSC & UAF

Study: Tanner crab genetic selection under severe low pH treatment

Ocean Acidification (OA) in the Bering Sea - model hindcast



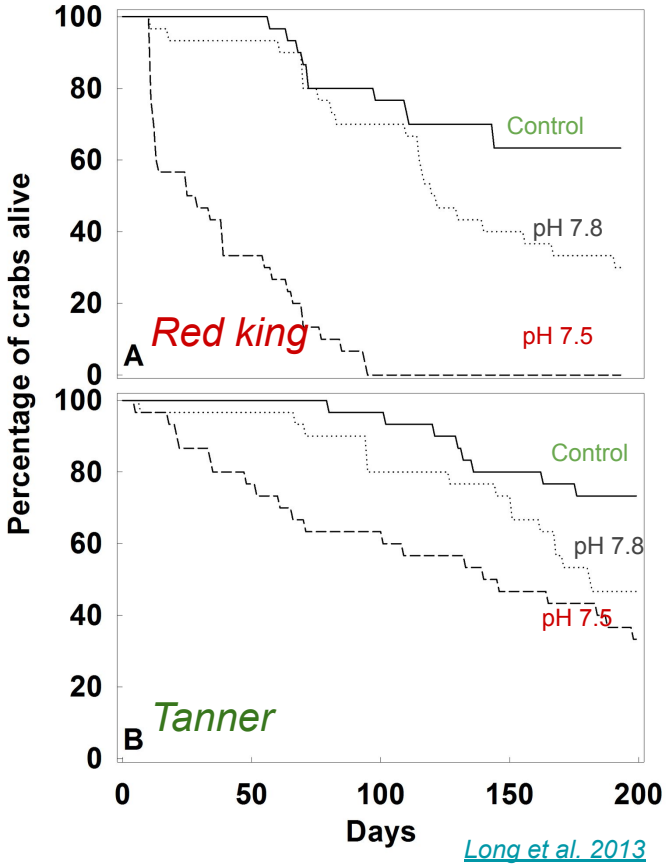
Pilcher et al. 2025



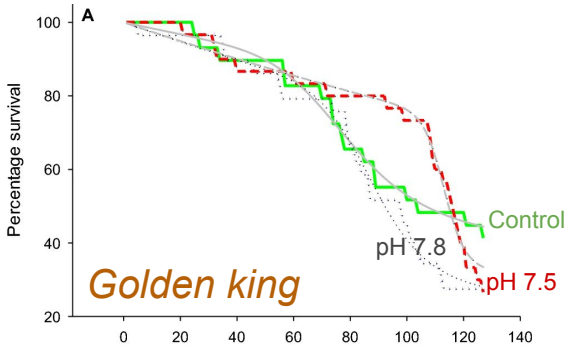
Amplified bottom water acidification rates on the Bering Sea shelf from 1970–2022

Darren J. Pilcher ✉, Jessica N. Cross, Natalie Monacci, Linquan Mu, Kelly A. Kearney, Albert J. Hermann, and Wei Cheng

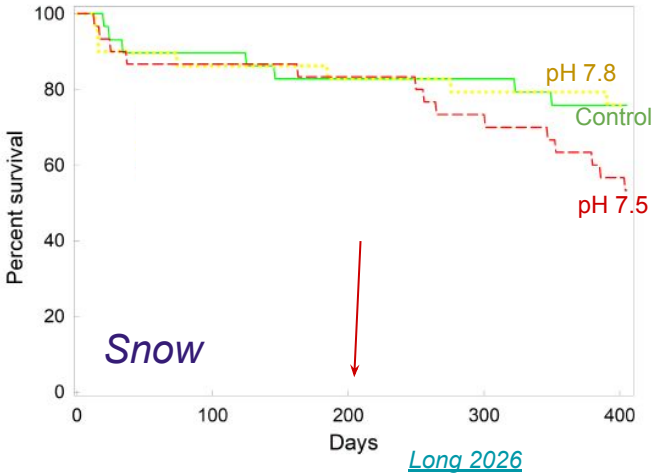
OA reduces crab survival, growth, but effects vary among species



Long et al. 2013



Long et al. 2021

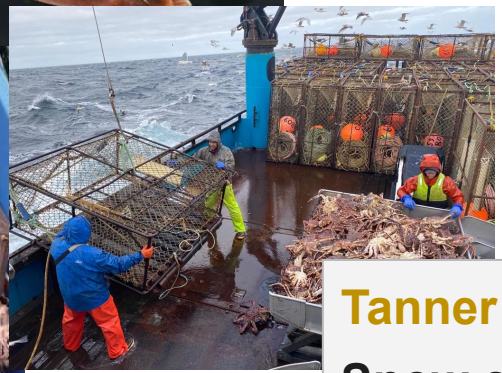


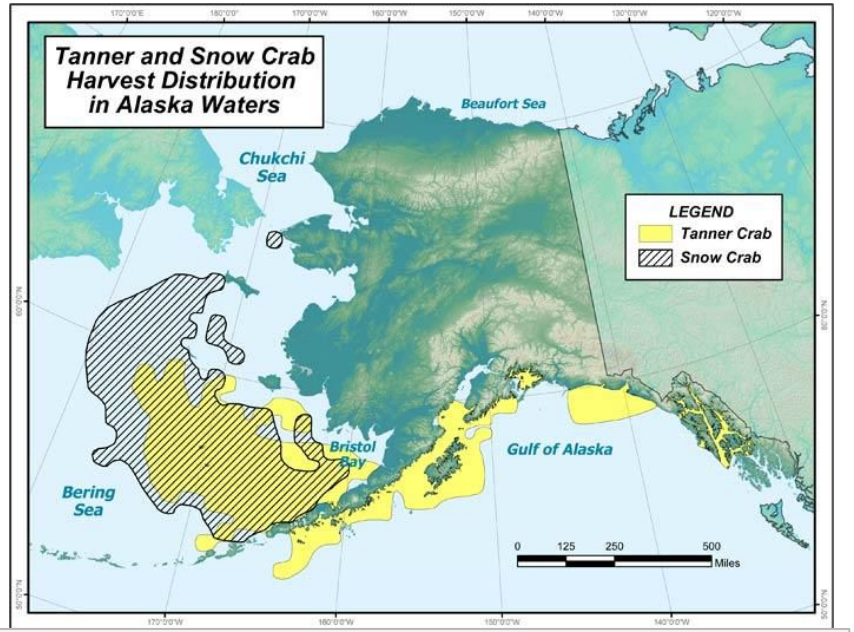
Long 2026

Tanner and snow crabs, *Chionoecetes spp.*





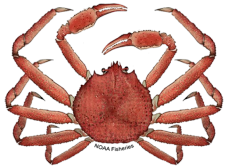




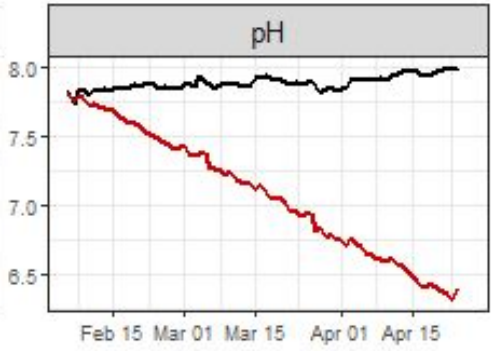
Tanner crab: **OA sensitive** **warm-tolerant**

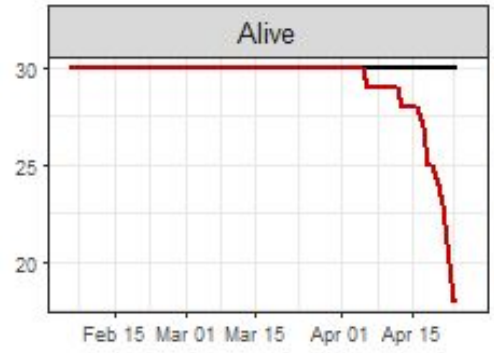
Snow crab: **OA tolerant** **warm-sensitive**

OA = ocean acidification OW = ocean warming



Experiment! What is distinct about Tanners that survive low pH stress?



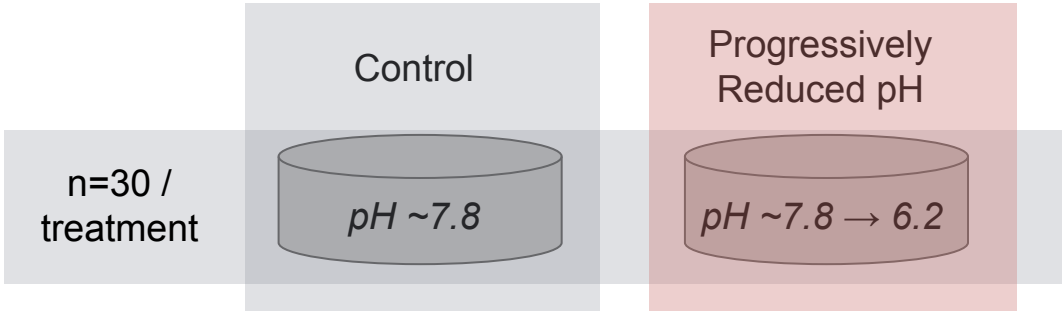


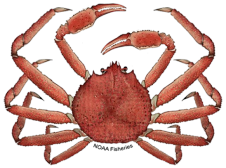
Laboratory Experiment

- 60 wild adult mature males from Eastern Bering Sea
- pH reduced over ~2 month until ~50% mortality

Data:

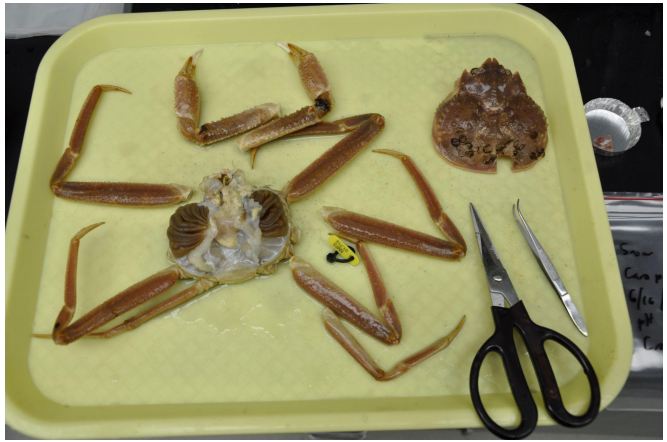
1. Survival, size
2. Molecular data





Data collected at experiment termination (50% mortality)

for all low pH survivors, subset of control group



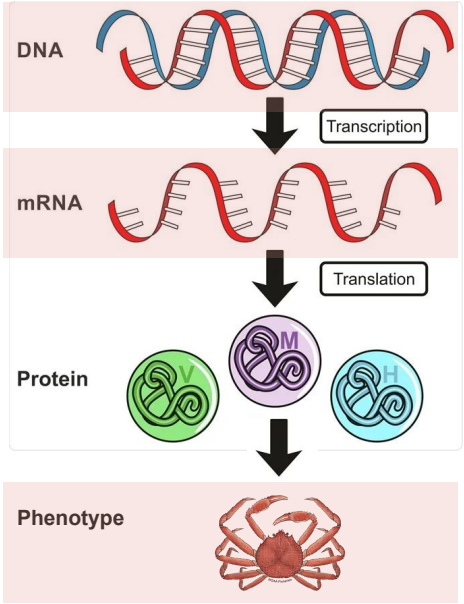
n=17 Control

n=14 low pH Survivors

Genetic data
- lcWGS, ~2s
- Same run/lane

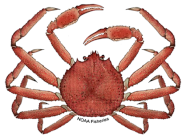
Expression data (RNAseq)
- Gill
- Hepatopancreas
- Muscle

Phenotypes
- Survival, Size



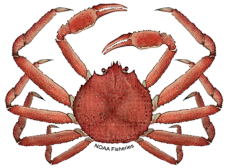
Adapted from: udaix/Shutterstock.com

What is distinct about Tanners that survive low pH stress?



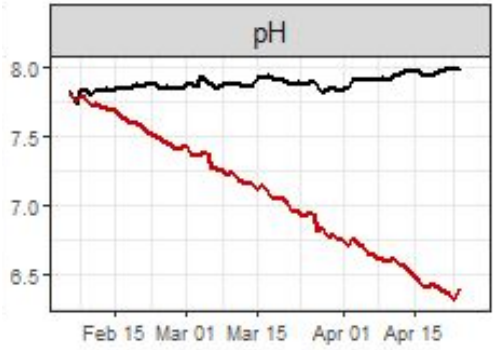
Question: Are there alleles/genotypes enriched in the Low pH Survivors?

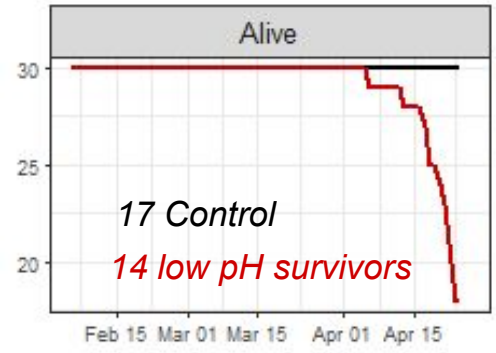
Feedback Please!

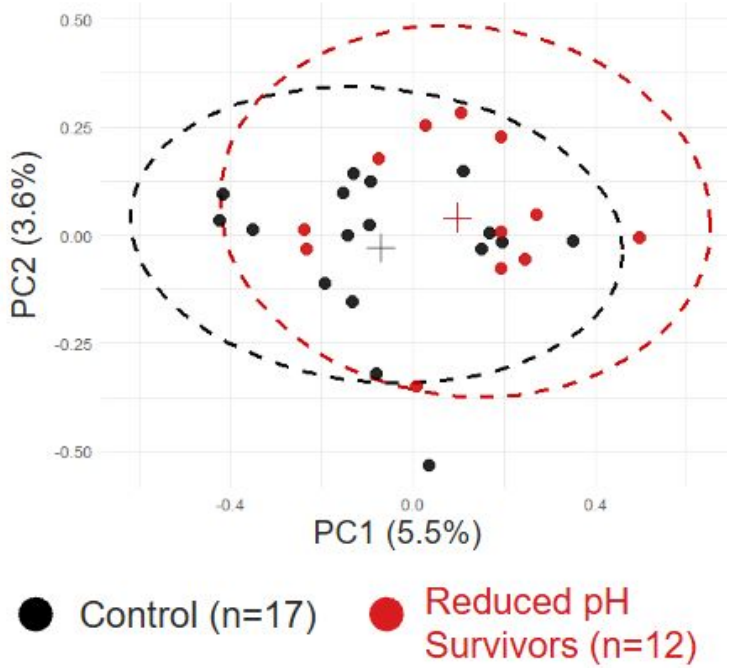


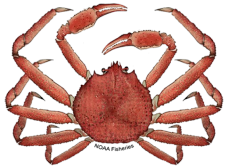
Subtle genome-wide clustering by treatment

PCA, all SNPs (~165k, pruned, depth=filtered)



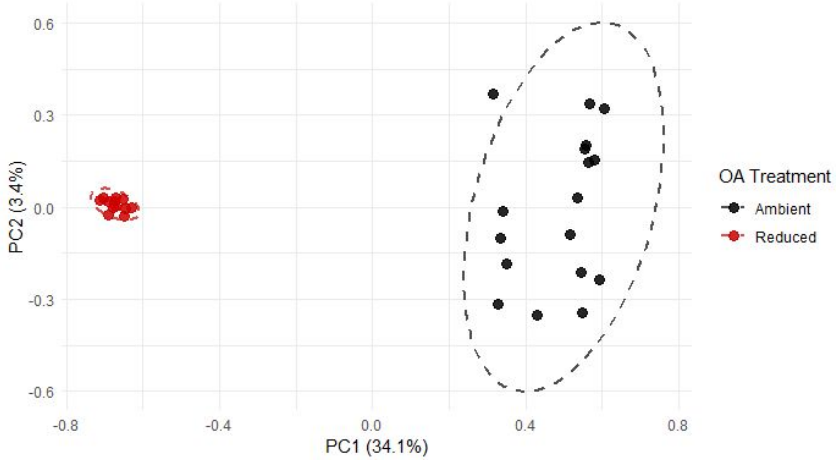




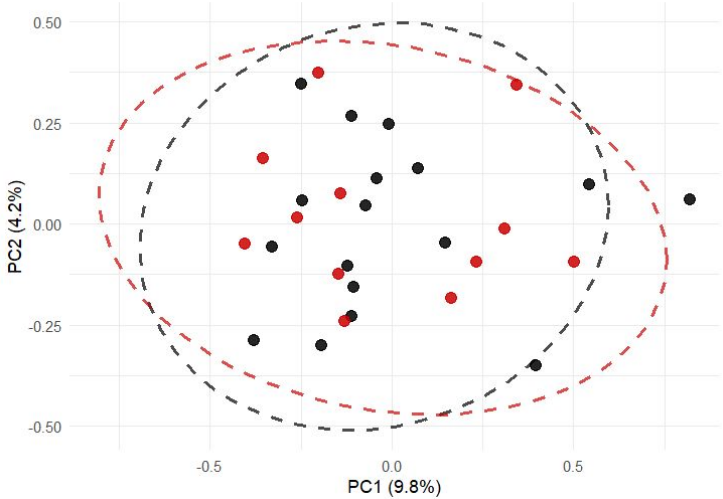


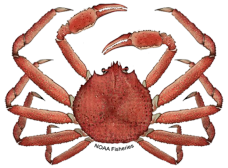
High FST sites → less variation in OA survivors

SNPs with Fst > 0.1 (unlinked, n=666)



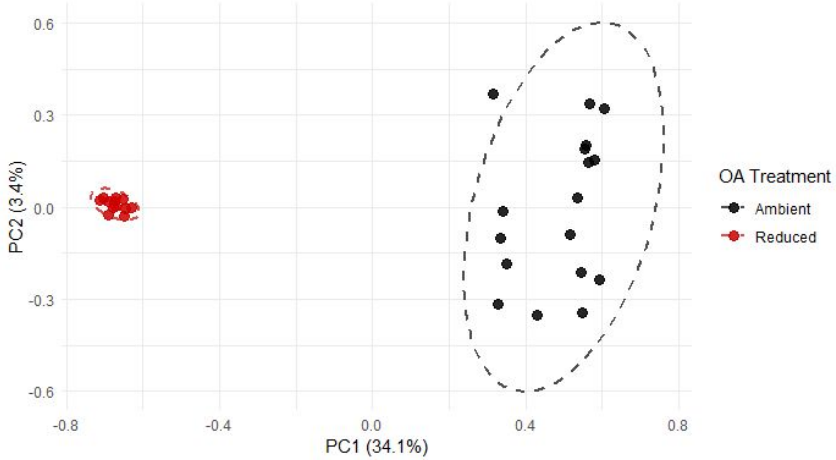
Random subset



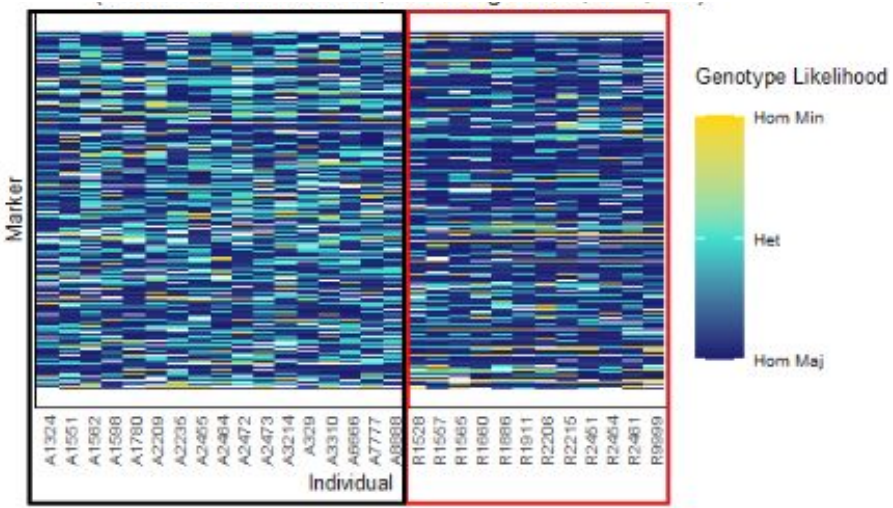


High FST sites → less variation in OA survivors
fewer heterozygotes

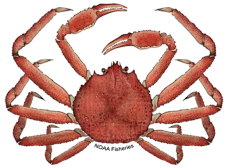
SNPs with Fst > 0.1 (n=666 unlinked)



SNPs with Fst > 0.1 (n=1,220, not pruned)



366 in annotated genes

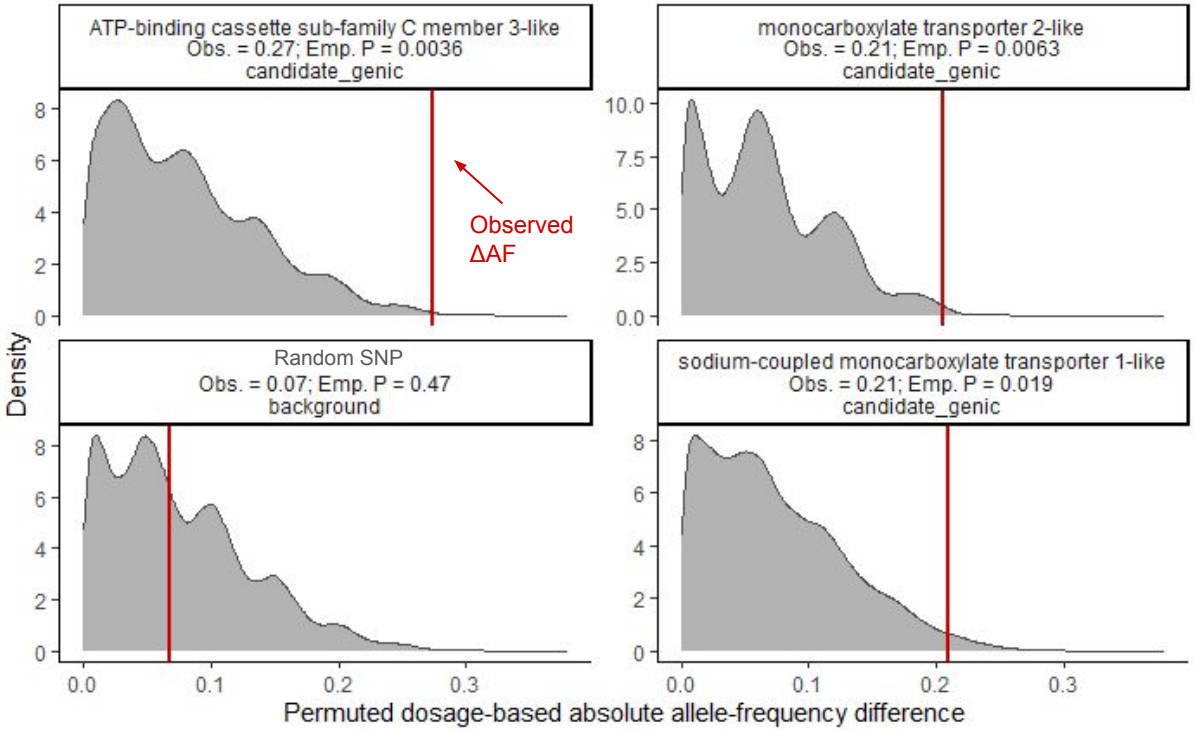


Allele frequency differences are higher than expected by chance

Null distributions from 100k permutations, samples/treatments randomized →

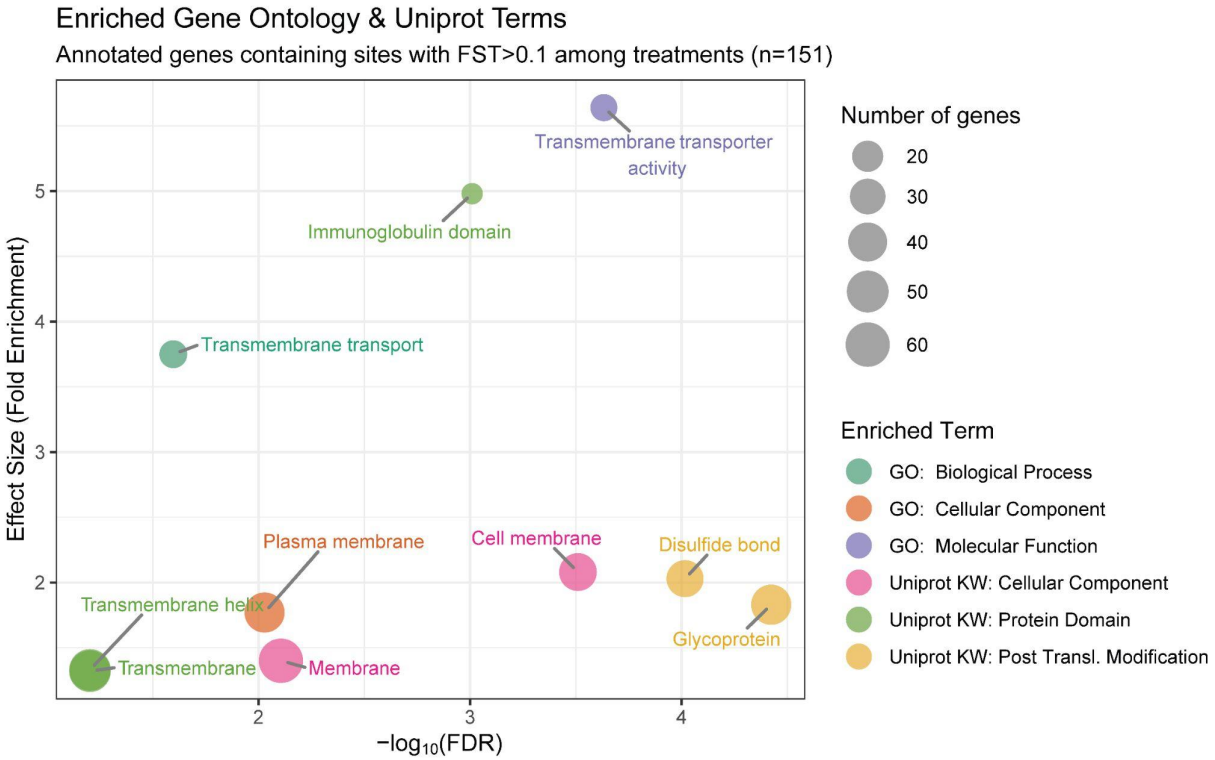
Conducted on candidate differentiated SNPs, + 50k random genes; 95% of SNPs > 95% confidence

Shown: 3 transmembrane SNPs + 1 random SNP

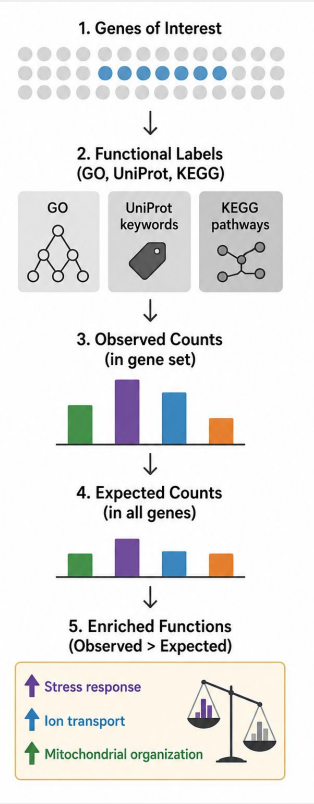




Genes containing these sites enriched for **transmembrane transport**

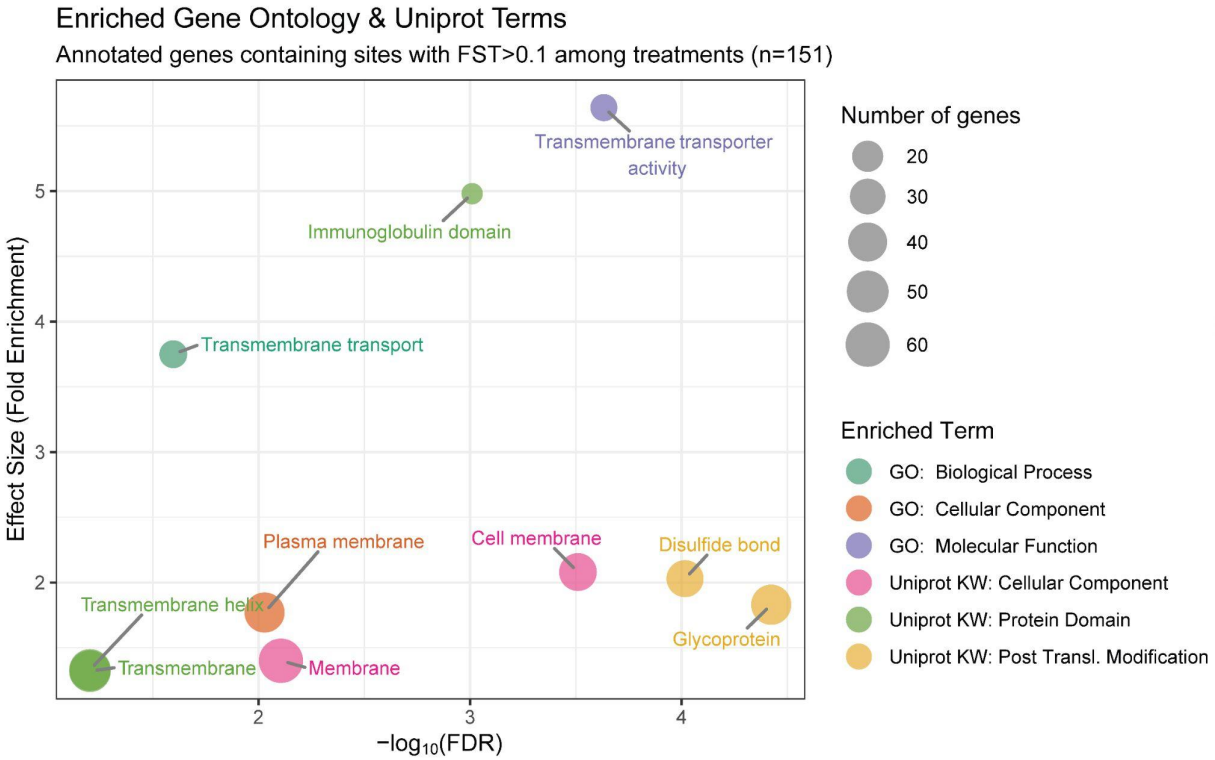


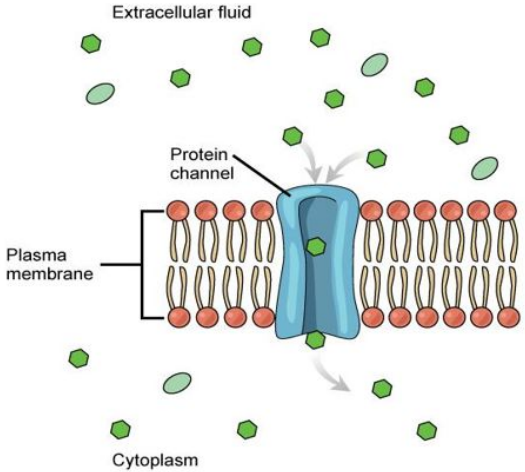
What is enrichment analysis?

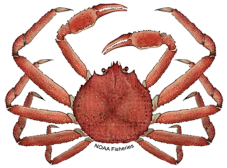




Genes containing these sites enriched for **transmembrane transport**

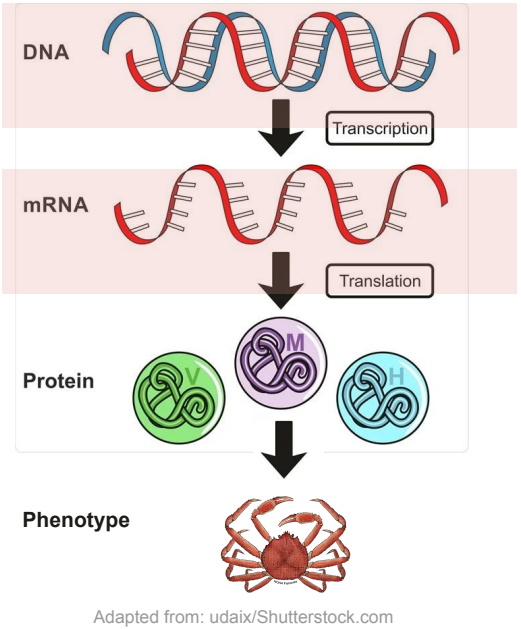




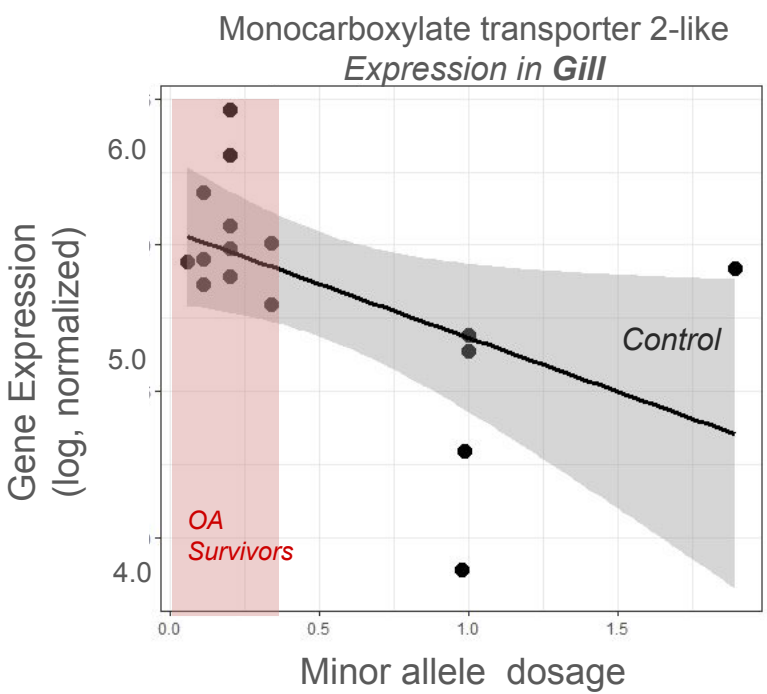


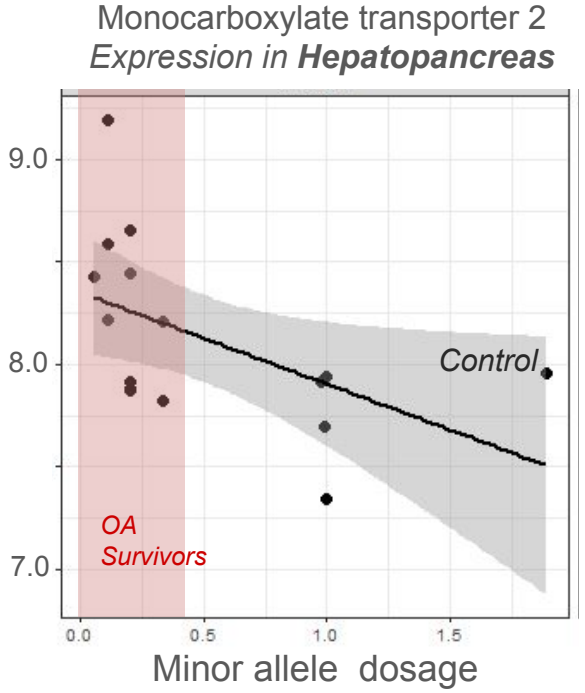
Some **genetic differences may influence gene expression**

(cis-eQTLs, SNPs within the same gene)



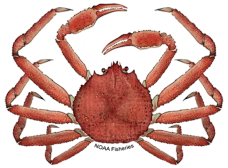
Adapted from: udaix/Shutterstock.com





$$\text{Dosage} = 0*P_{maj} + 1*P_{het} + 2*P_{min}$$

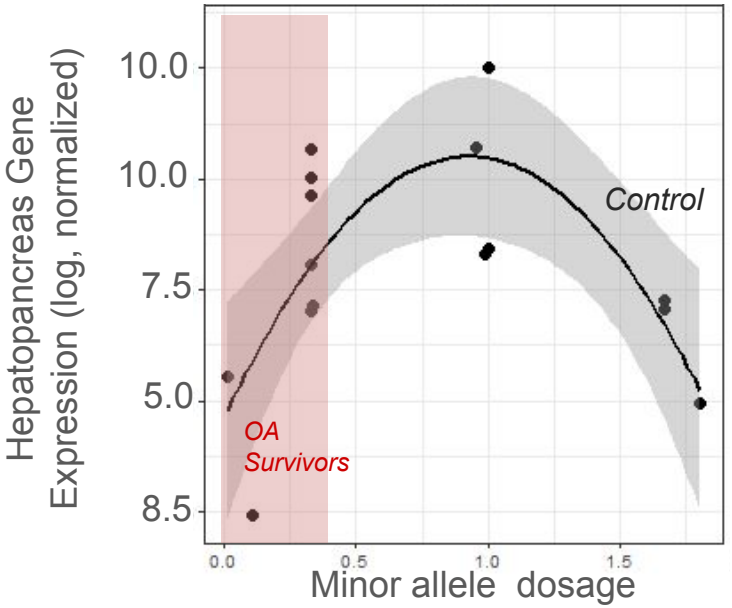
Proton-coupled symporter. Rapid transport of energy substrates (e.g. lactate) across cell membranes (gene name: SLC16A7)



Some **genetic differences may influence gene expression**

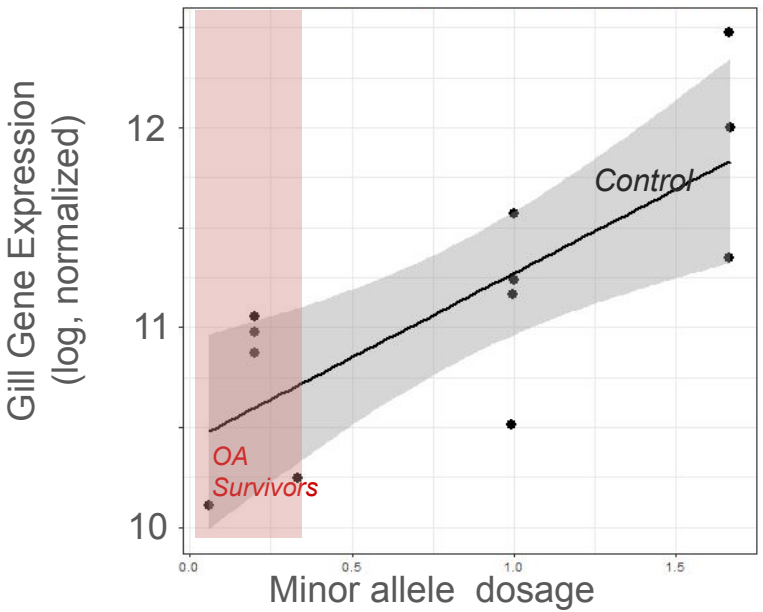
(cis-eQTLs, SNPs within the same gene)

ATP-binding cassette sub-family C member 3



Efflux pump. Transports organic anions, bile salts, drugs, toxicants, cross cell membranes

Heme peroxidase 2



Immunity, defense. Maintains cuticle integrity in the hypodermis

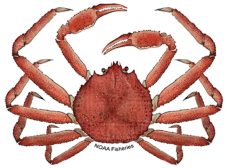
What is distinct about Tanners that survive in OA?



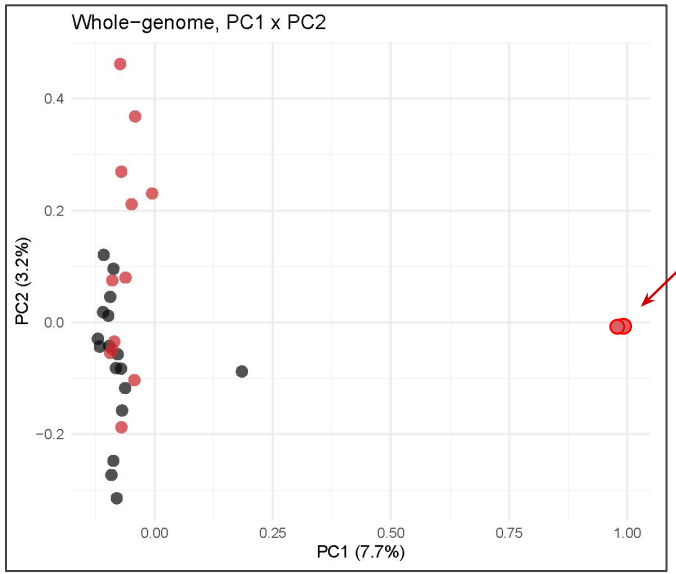
1. **Low pH survivors have less genetic variation** (fewer heterozygotes, minor alleles) where there are genetic differences

2. OA tolerance may be associated with **effective transmembrane transport pathways**

One more thing …



Genetic outliers, who are they?



????

OA Treatment

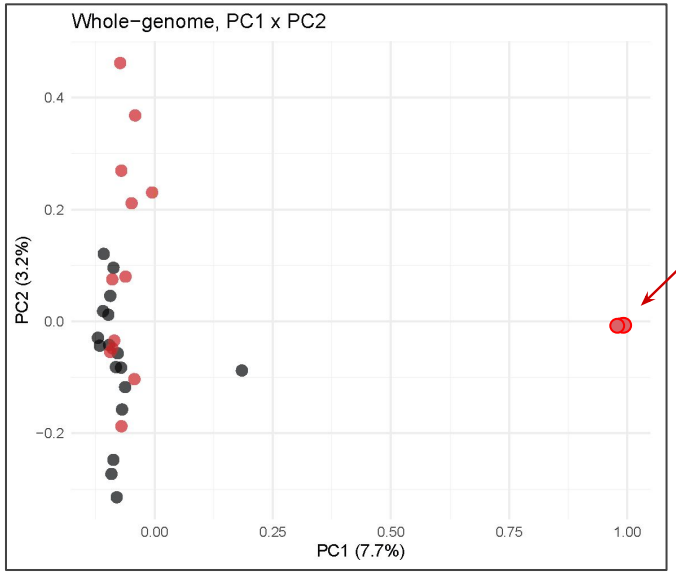
Ambient

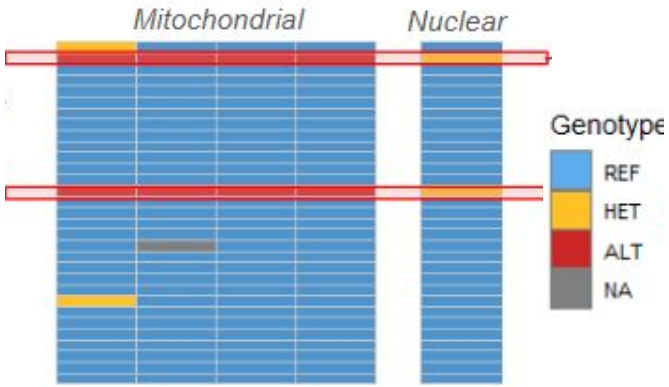
Reduced

Genetic outliers, who are they?

Snow-Tanner hybrids!

Two of 14 low pH survivors





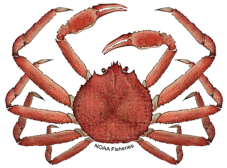
SNP that differ among tanner crab (REF) and snow crab (ALT)

A Rapid, High-Throughput Technique for Detecting Tanner Crabs *Chionoecetes bairdi* Illegally Taken in Alaska's Snow Crab Fishery Get access

Christian T. Smith, W. Stewart Grant, Lisa W. Seeb

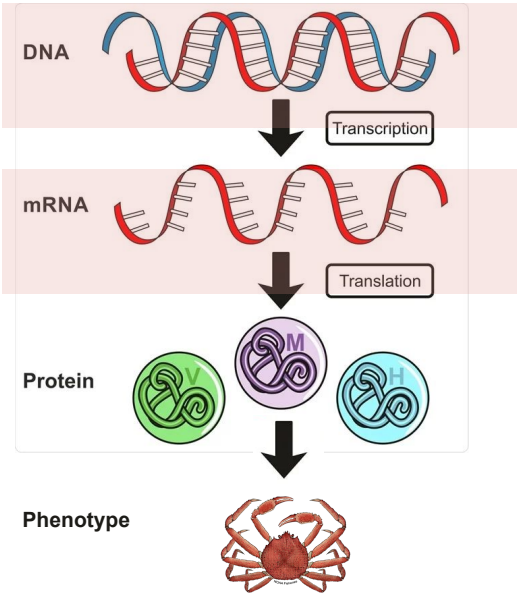
Transactions of the American Fisheries Society, Volume 134, Issue 3, May 2005, Pages 620–623, https://doi.org/10.1577/T04-007.1

Published: 01 May 2005 **Article history**

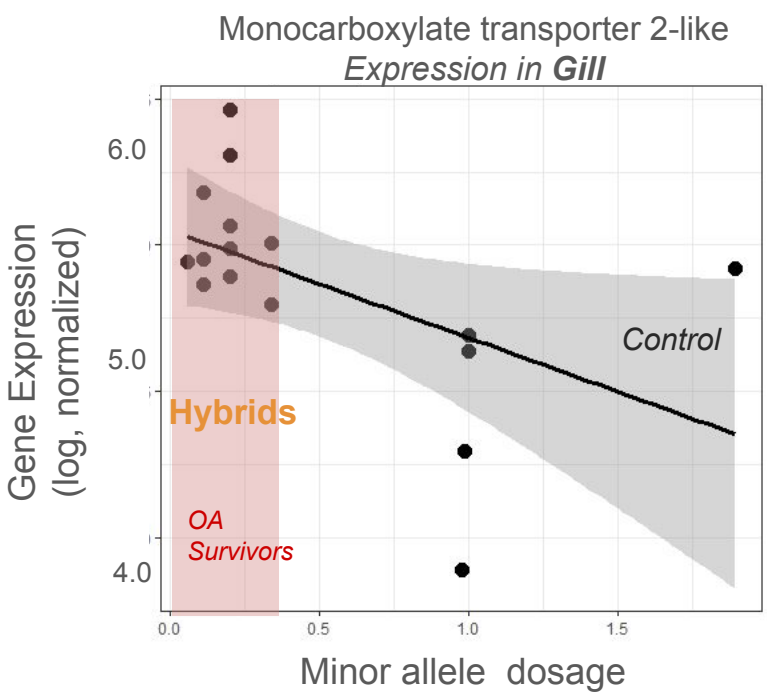


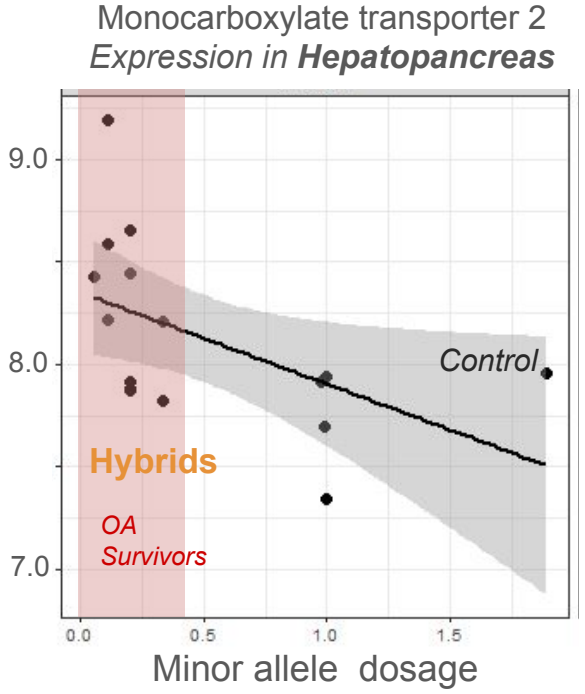
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(cis-eQTLs, SNPs within the same gene)



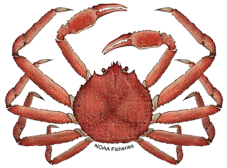
Adapted from: udaix/Shutterstock.com



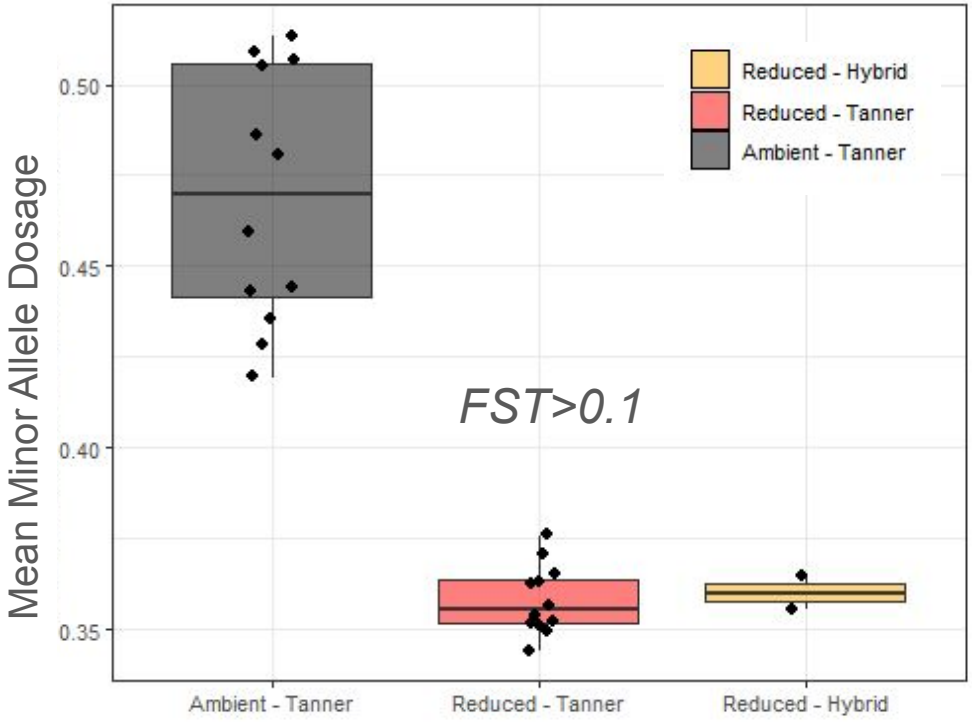


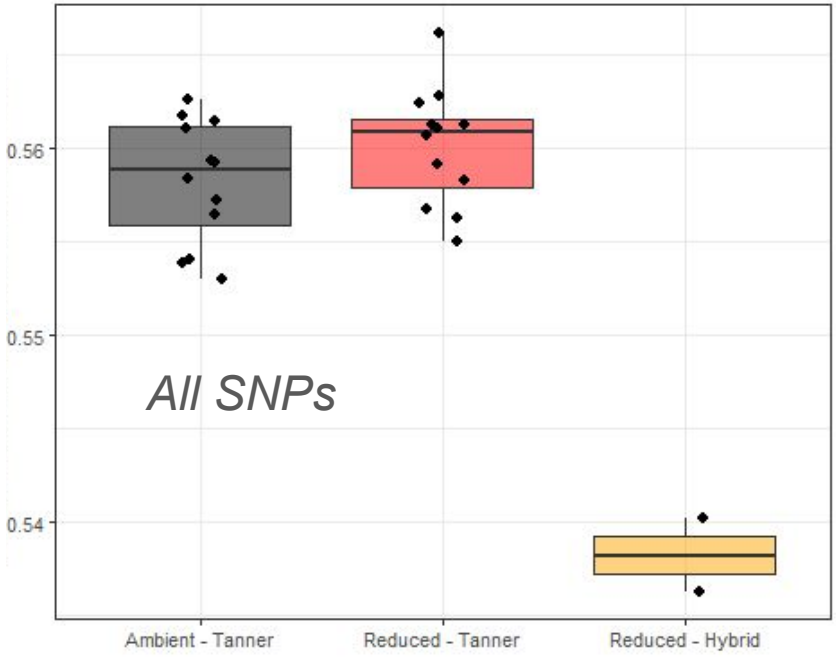
$$\text{Dosage} = 0*P_{maj} + 1*P_{het} + 2*P_{min}$$

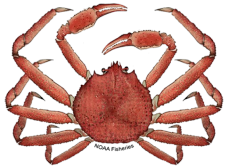
Proton-coupled symporter. Rapid transport of energy substrates (e.g. lactate) across cell membranes (gene name: SLC16A7)



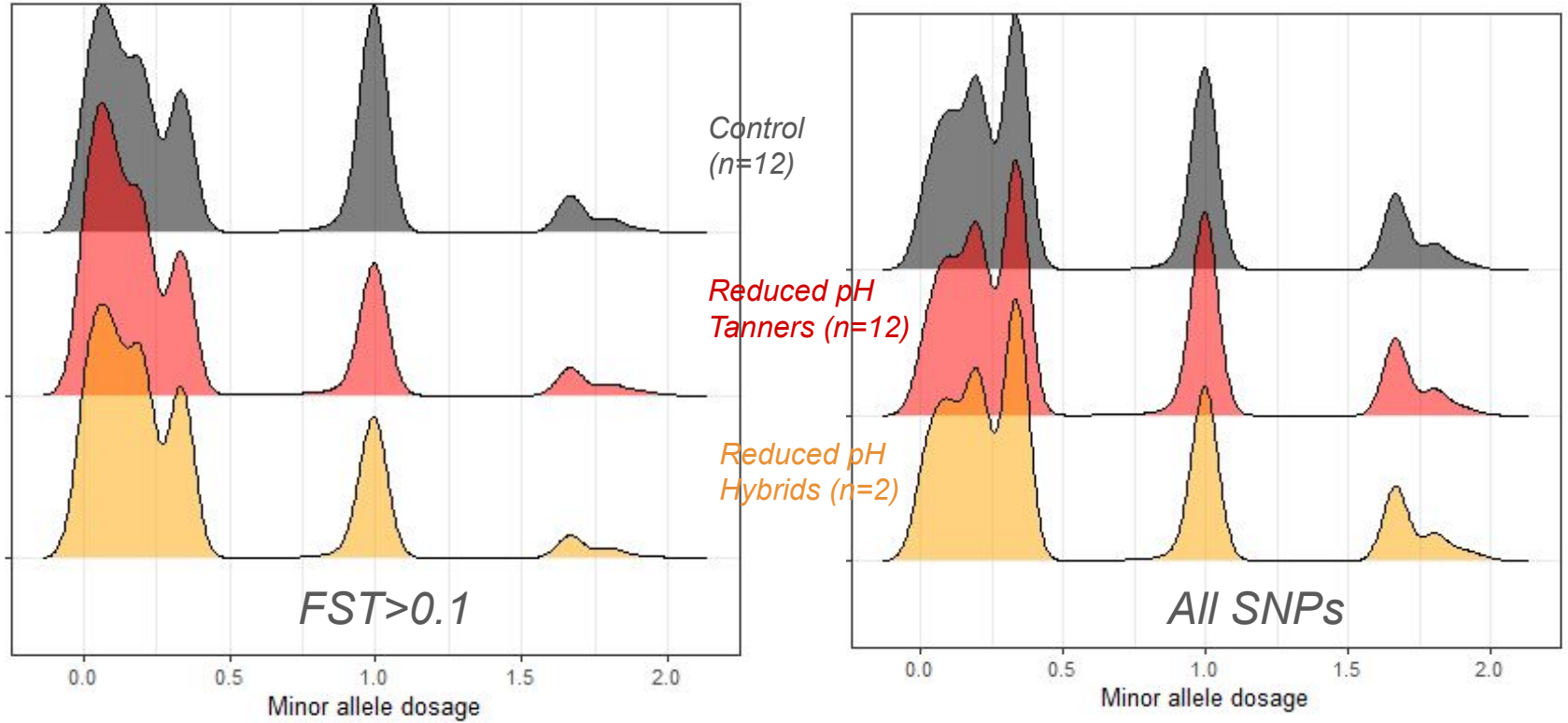
Hybrids & Tanners that survived reduced pH have lower minor allele dosage at differentiated sites

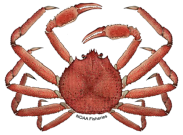






Hybrids & Tanners that survived reduced pH have lower minor allele dosage at differentiated sites





Next Steps

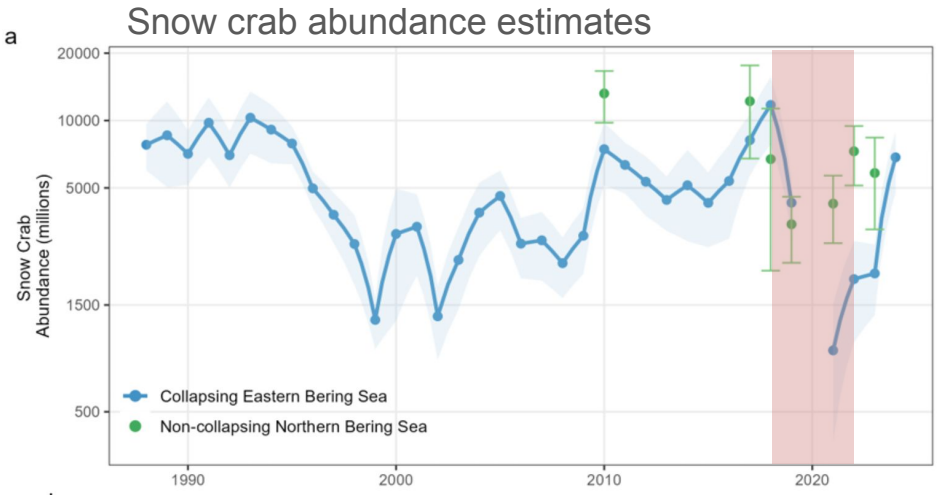
1. Differential Exon Usage ~ Minor Allele Dosage (euQTLs)
2. Non-synonymous mutation analysis – do SNPs alter amino acid sequences at QTLs?

Study limitations

1. No genetic data from morts!
2. lcWGS data is noisy
3. Low sample sizes, limited power
4. Genome
5. Only two hybrids

Study: Exploring Snow crab OA tolerance, and limitations, using gene expression

Snow crab population changes linked to environmental conditions



Fedewa, Copeman, & Litzow 2025

Energetic limitations and mass mortality of Bering Sea snow crab: Interacting effects of warming and density on collapse and recovery

Authors: Erin J. Fedewa, Louise A. Copeman, and Michael A. Litzow | AUTHORS INFO & AFFILIATIONS

Publication: Canadian Journal of Fisheries and Aquatic Sciences • 24 November 2025 • https://doi.org/10.1139/cjfas-2025-0099

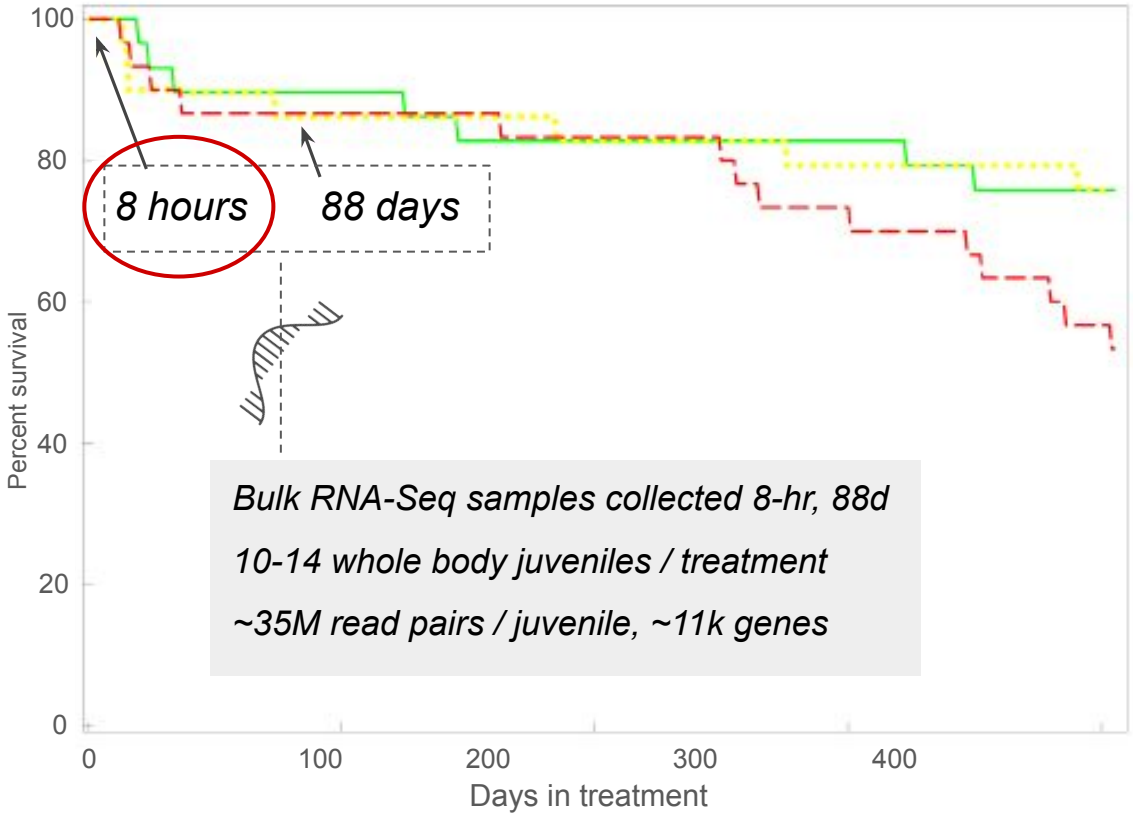
EBS snow crab mass mortality during heatwaves, likely due to starvation, first EBS snow crab fishery closure in history

Recruitment is key to recovery!

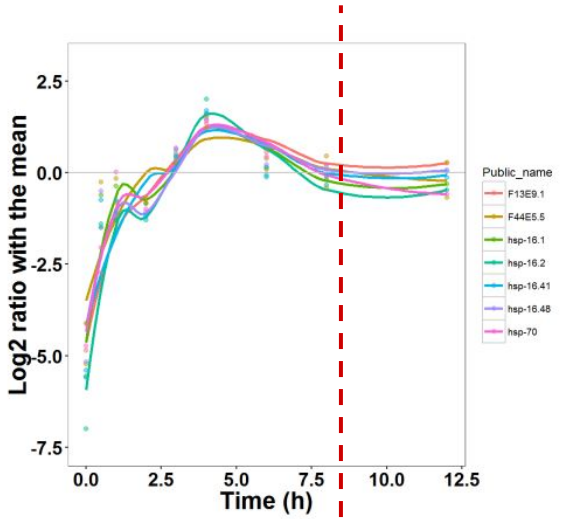
Early life stages sensitive to environmental stressors

Does ocean acidification affect recruitment?

Gene expression analysis used to understand snow crab OA tolerance & limits

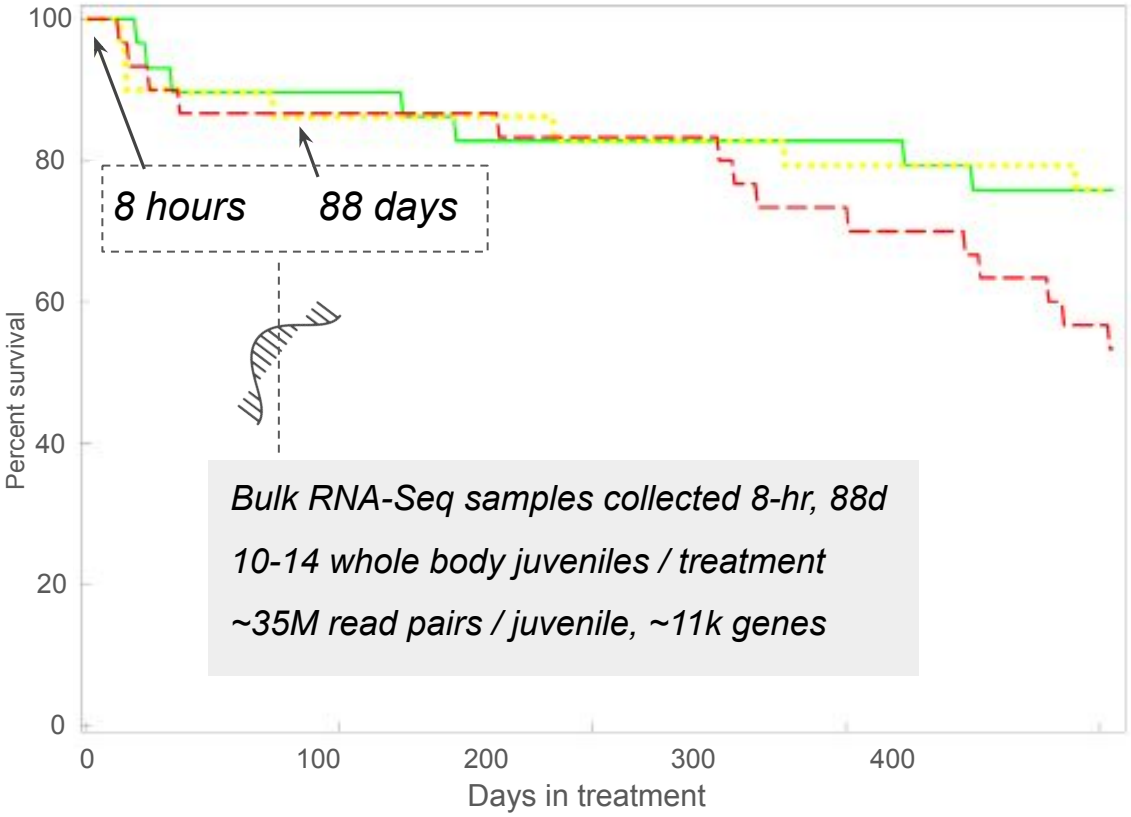


8 hr: Capture expression patterns during immediate acclimation to new environment



C. elegans expression in continuous heat stress
Jovic et al. 2017

Gene expression analysis used to understand snow crab OA tolerance & limits



Typical workflow

Sequence all transcripts (mature mRNA)

↓

Trim/filter reads *Cutadapt*

↓

Alignment to snow reference genome, *STAR*

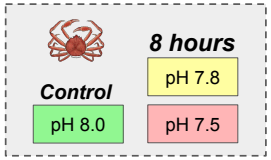
↓

Gene counts (~11K genes) *STAR, or featureCounts*

PCA

Diff. gene expression *DESeq2*

WGCNA DTU/DEU



Mitochondrial pathways activated during immediate OA acclimation

Higher expression in OA

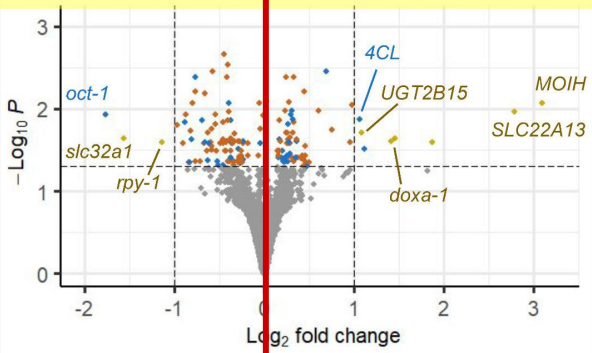
Lower Expression in OA

Higher Expression in OA

n=87 DEGs

No enriched processes

8-hr acclimation to pH 7.8



n=61 DEGs

Mitochondrial electron transport

Oxidoreductase activity

22 shared DEGs

No enriched processes

19 shared DEGs

Mitochondrial activity

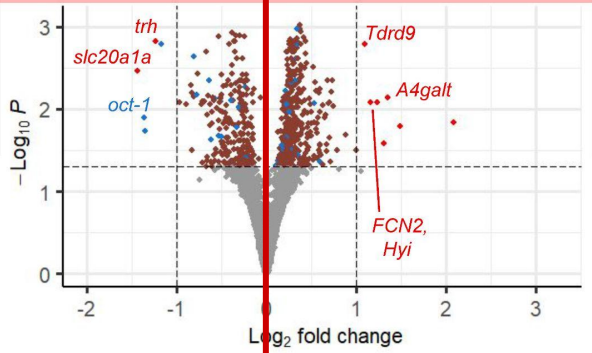
n=240 DEGs

Cell adhesion

Lipid transport

Endoplasmic reticulum stress response

8-hr acclimation to pH 7.5

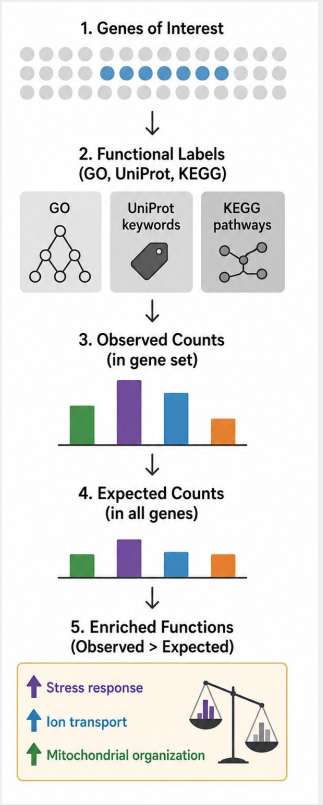


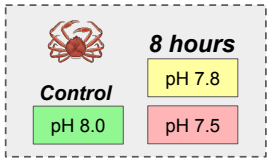
n=376 DEGs

Mitochondrial electron transport, translation

Respiratory chain complex I assembly

What is enrichment analysis?





Mitochondrial pathways activated during immediate OA acclimation

Higher expression in OA

Lower Expression in OA

Higher Expression in OA

n=87 DEGs

No enriched processes

8-hr acclimation to pH 7.8

n=61 DEGs

Mitochondrial electron transport

Oxidoreductase activity

22 shared DEGs

No enriched processes

19 shared DEGs

Mitochondrial activity

n=240 DEGs

Cell adhesion

Lipid transport

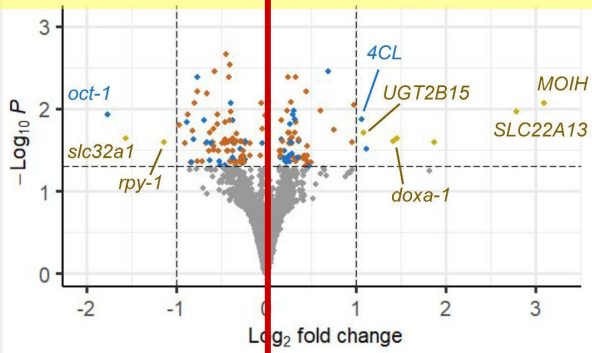
Endoplasmic reticulum stress response

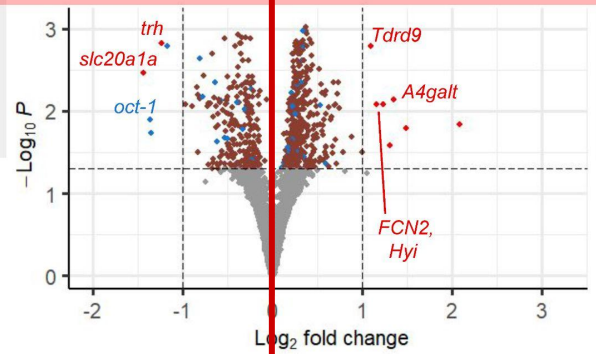
8-hr acclimation to pH 7.5

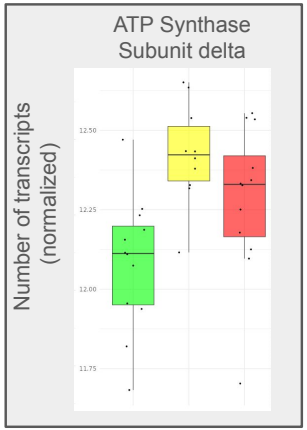
n=376 DEGs

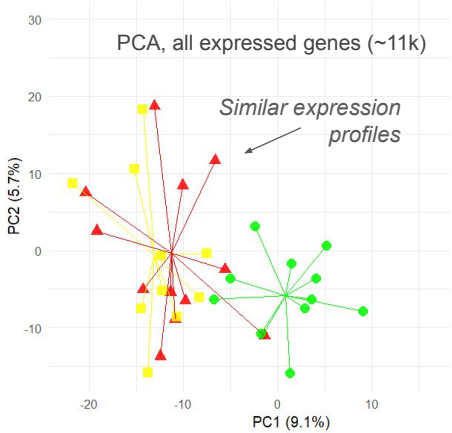
Mitochondrial electron transport, translation

Respiratory chain complex I assembly

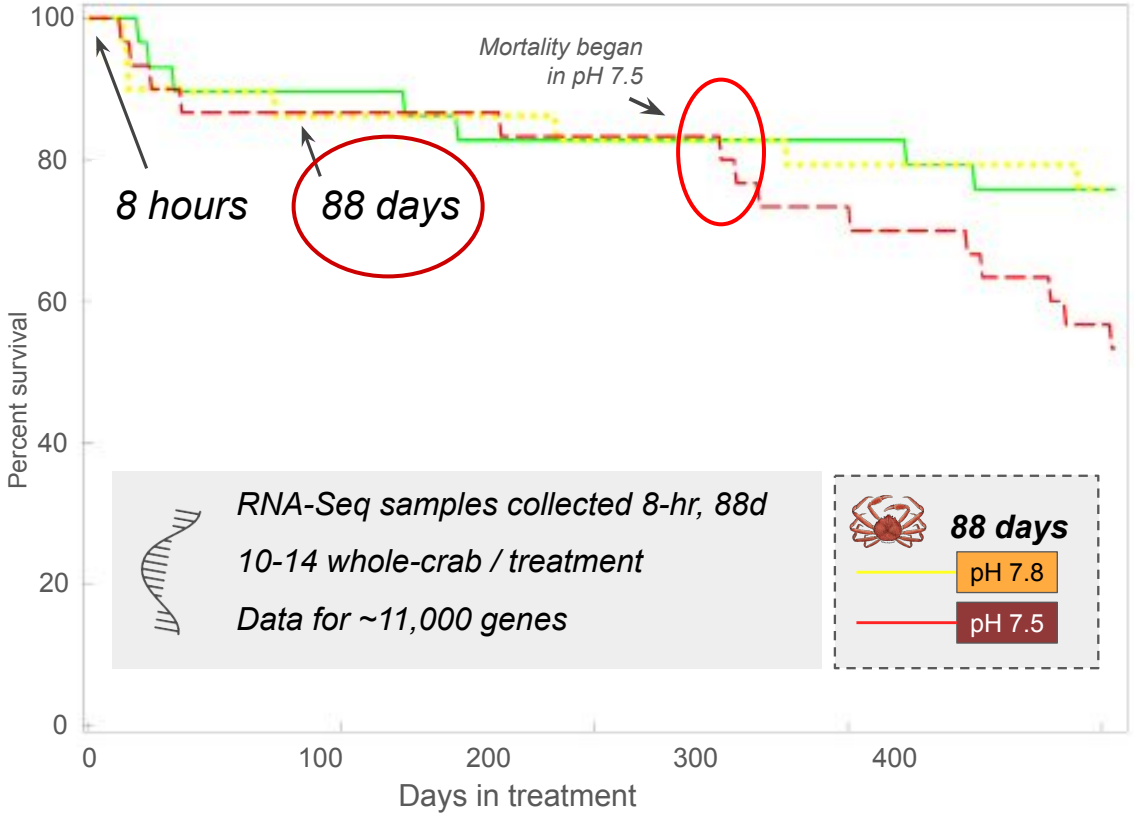




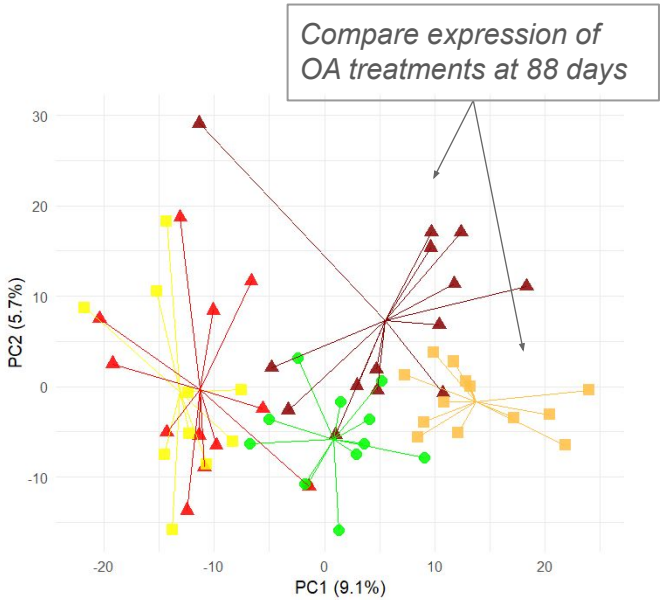


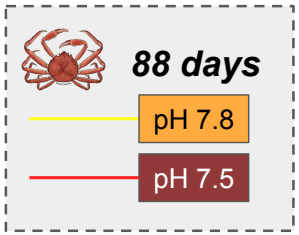


Gene expression analysis used to understand snow crab OA tolerance & limits

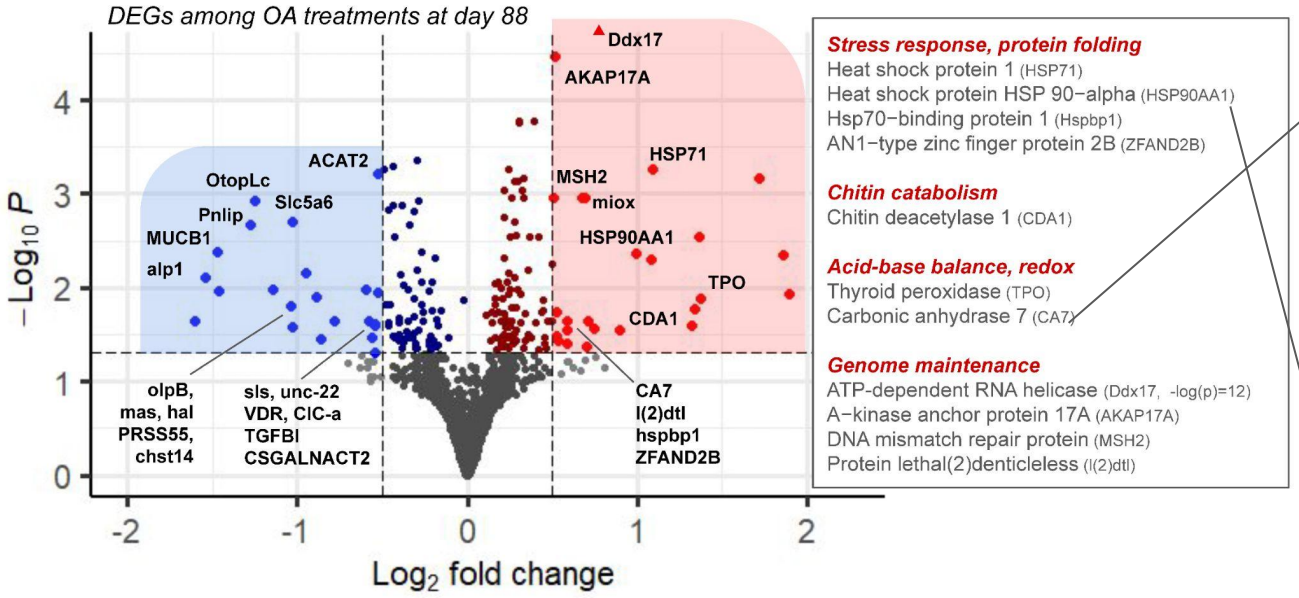


Note: No control at day 88 :(





Higher expression of damage response, pH regulation, cuticle maintenance in severe OA - indicative of stress

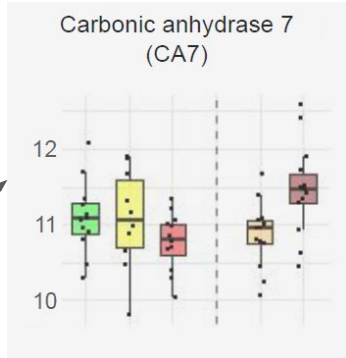


Stress response, protein folding
Heat shock protein 1 (HSP71)
Heat shock protein HSP 90−alpha (HSP90AA1)
Hsp70−binding protein 1 (Hspbp1)
AN1−type zinc finger protein 2B (ZFAND2B)

Chitin catabolism
Chitin deacetylase 1 (CDA1)

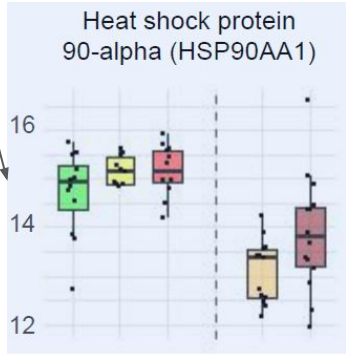
Acid-base balance, redox
Thyroid peroxidase (TPO)
Carbonic anhydrase 7 (CA7)

Genome maintenance
ATP-dependent RNA helicase (Ddx17, -log(p)=12)
A−kinase anchor protein 17A (AKAP17A)
DNA mismatch repair protein (MSH2)
Protein lethal(2)denticleless (l(2)dtl)

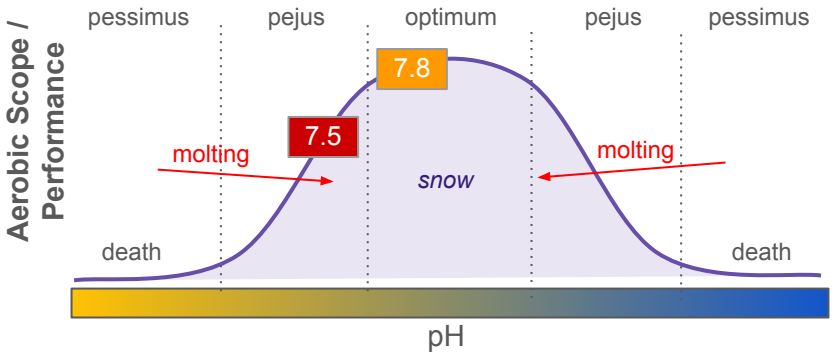


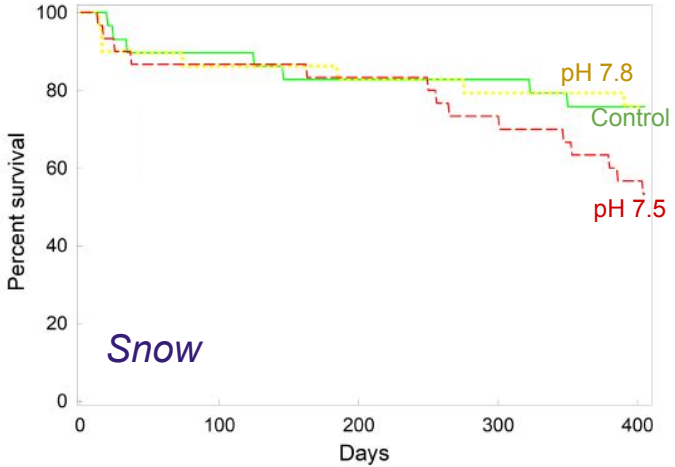
Top candidate biomarker of OA stress

$CO_2 + H_2O \rightleftharpoons$ bicarbonate + H^+

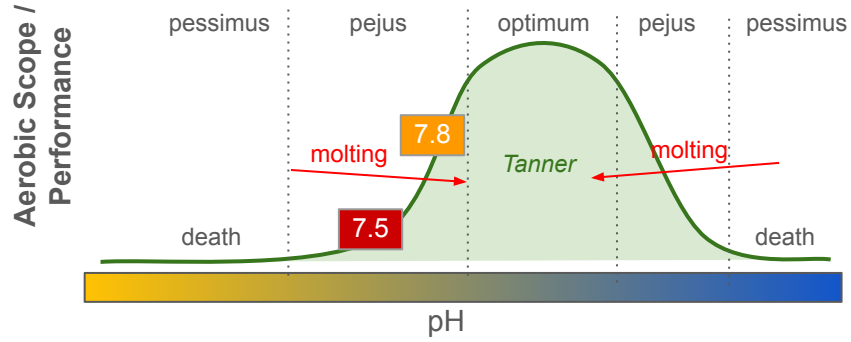


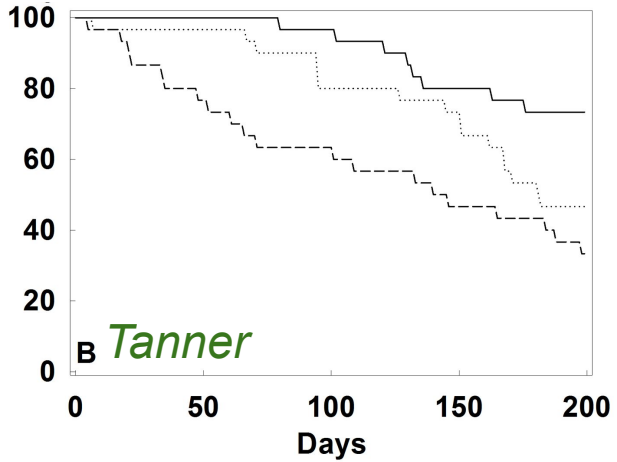
High expression on Day 1 - handling? New lab environment?



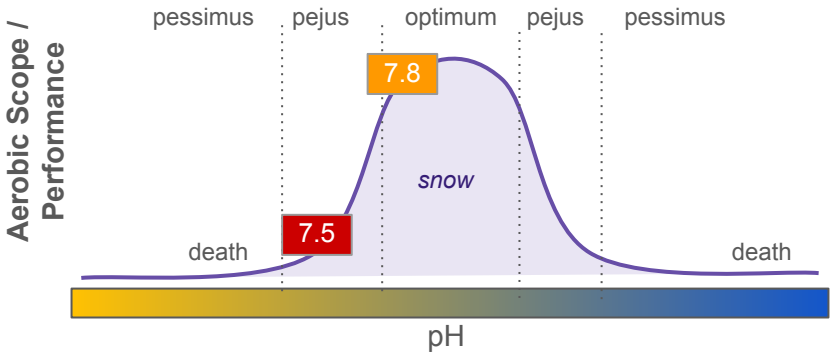


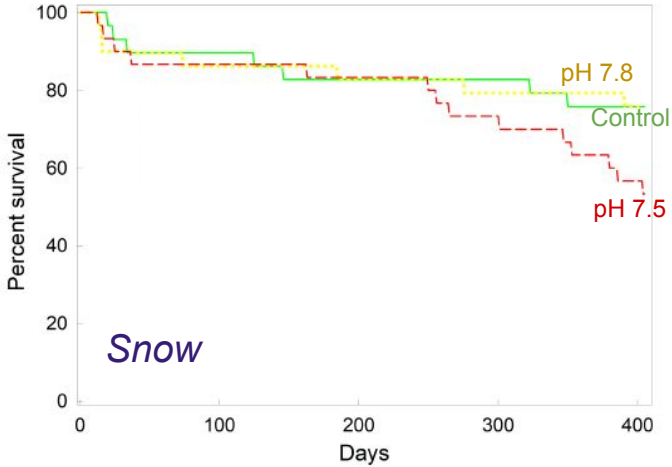
[Long 2026]



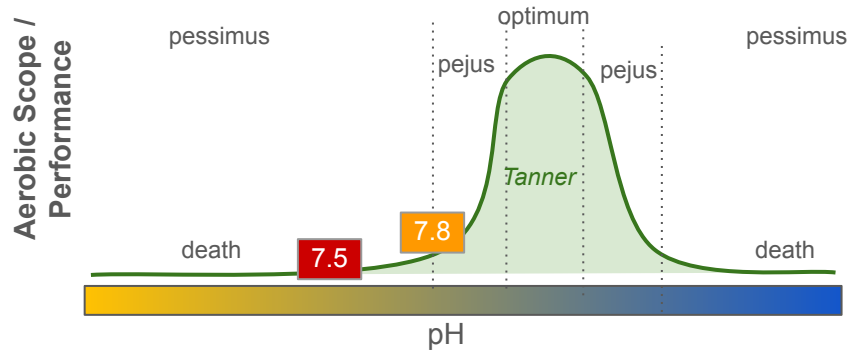


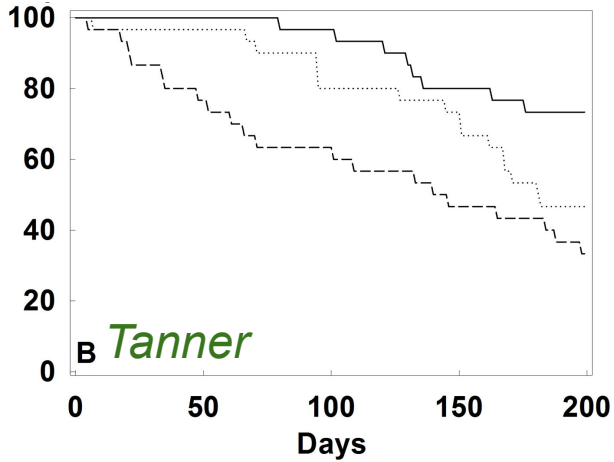
[Long et al. 2013]



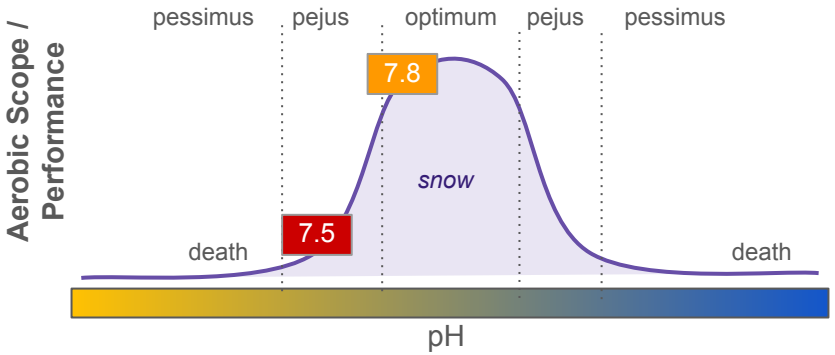


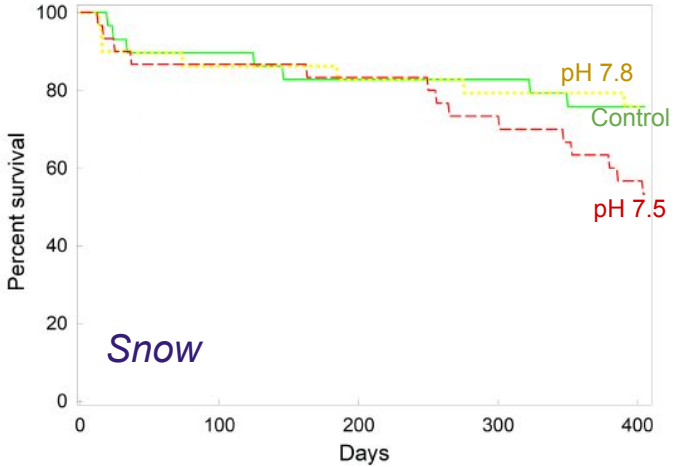
[Long 2026]





[Long et al. 2013]

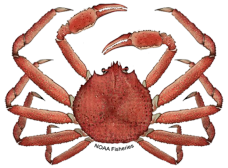




Long 2026

Future needs, steps, ideas

- Better genome annotation ← *I made a new one! Via EGAPx (NCBI pipeline)*
- Tissue specific expression, direct comparison to Tanners - *Next project, data in-hand*
- Test CA7 in distinct tissues, and range of conditions – is CA7 a reliable indicator of low pH stress?
- Metabolic assays (e.g. expression, O_2 consumption) across range of pH, time points (temperature too!)



Short-term mechanisms, long term consequences

Snow crab juveniles are unaffected by moderate OA (pH 7.8), but more severe OA (pH 7.5) reduces survival, beginning at 250 days exposure.

Short-term OA tolerance mechanisms …

- Investment in energy production, repair in mitochondria

Long-term OA tolerance consequences …

- Chronic cellular stress mitigation (protein, DNA damage) & chitin maintenance
- Mortality during molting - over-extended stress/damage-response systems

Snow crab OA gene expression explorer

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